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Theoretical Physics - Complex Networks

Criticality in Epilepsy: A Topological and Functional Analysis of Intracranial Recordings

Final Project

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Abstract

The transition from a physiological inter-ictal state to an epileptic seizure is a complex dynamical process characterized by abrupt shifts in cerebral synchronization. This work investigates the pre-ictal to ictal transition in focal epilepsy using human stereo-electroencephalography (sEEG) recordings analyzed through the framework of Complex Network Theory and statistical mechanics. By constructing time-varying functional networks via zero-lag Pearson correlation without absolute value rectification ($W_{ij}(\tau) \in [-1, 1]$), we strictly separate cooperative in-phase synchronization from competitive counter-phase decoupling. Our multiscale methodology tracks the macroscopic percolation phase transition and topological susceptibility of the network, the mesoscopic collapse of modular community structures via Newman's modularity and the Louvain heuristic, and the microscopic functional reorganization of individual recording contacts. By evaluating node metrics across robust statistical epochs rather than noisy instantaneous frames, we introduce a two-dimensional Functional Role Map that classifies recording channels into premonitory initiators, secondary ictal amplifiers, and counter-phase poles. Finally, we demonstrate that purely topological network descriptors can effectively capture the critical symmetry breaking and functional architecture of the epileptogenic network, independent of three-dimensional Euclidean spatial coordinates.

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1 | Introduction and Physical-Clinical Framework

Epilepsy has historically been conceptualized as a focal disorder originating from an isolated, hyperexcitable cortical region. Over the past two decades, however, clinical neurophysiology and computational neuroscience have undergone a paradigm shift, redefining focal epilepsy as a large-scale network disorder [1, 2]. Seizure generation (ictogenesis) and propagation do not merely depend on the intrinsic excitability of a single anatomical locus, but emerge from the collective dynamical interactions among distributed neural populations.

Understanding the mechanisms that drive the sudden transition from physiological background activity to a full-blown seizure requires moving beyond the analysis of single-channel voltage time series. By adopting the mathematical tools of statistical mechanics and graph theory, the brain can be modeled as a complex network of coupled nonlinear oscillators, allowing us to characterize an epileptic seizure as a critical phase transition across multiple topological scales [3, 4].

1.1 | The Brain as a Critical Complex System

The human cerebral cortex is a quintessential complex system composed of billions of neurons interacting through excitatory and inhibitory synaptic pathways across multiple spatial and temporal scales. Under healthy physiological conditions, cortical networks must simultaneously satisfy two competing functional demands: *functional segregation*, which enables specialized information processing within localized neuronal assemblies, and *functional integration*, which coordinates distributed processes into coherent cognitive states [4].

A growing body of theoretical and empirical evidence in statistical physics suggests that the healthy brain achieves this optimal trade-off by operating near a critical point—on the boundary of a second-order phase transition between an ordered (quiescent or hypersynchronous) phase and a disordered (random, asynchronous) phase [5, 6]. Operating in this critical regime maximizes the system's dynamic range, susceptibility to external inputs, and information storage capacity, while preventing runaway excitation through a delicate homeostatic balance between synaptic excitation and inhibition.

From the perspective of dynamical systems and statistical mechanics, focal epileptic seizures represent a catastrophic breakdown of this homeostatic criticality, driving the cortical network across a bifurcation into a pathological, supercritical regime of global hypersynchronization [2, 7]. Clinically, the temporal evolution of a seizure recording is partitioned into distinct dynamical states:

- **Inter-ictal (Baseline) State:** Far from a seizure, neuronal populations exhibit low-amplitude, high-complexity fluctuations. Functional interactions remain largely localized within specialized anatomical modules, preventing macroscopic runaway synchronization.
- **Pre-ictal Transition State:** In the seconds to minutes immediately preceding seizure onset, subtle dynamical shifts destabilize the baseline network. Crucially, neurophysiological studies have demonstrated that seizure onset is frequently preceded not by a gradual build-up of global synchronization, but by a paradoxical *pre-ictal desynchronization* or active functional decoupling [8, 7]. During this premonitory window, contacts within the putative Seizure Onset Zone (SOZ) functionally detach from the surrounding healthy cortex, often oscillating in counter-phase (anticorrelation) as local inhibitory restraints begin to fail.
- **Ictal Onset and Propagation:** At the critical transition point ($t = 0$), the decoupled pathological core overcomes surrounding inhibitory barriers—a phenomenon known clinically as the breakdown of the "inhibitory restraint"—and abruptly recruits adjacent and distant brain regions into high-amplitude, in-phase synchronous discharges [1]. In physical terms, the system undergoes spontaneous symmetry breaking, collapsing from a high-dimensional, fragmented state into a low-dimensional,

macroscopically ordered phase.

Investigating these rapid transitions in human patients requires recording modalities with exceptional temporal and spatial resolution. Non-invasive techniques such as scalp EEG suffer from severe spatial smearing and attenuation due to the skull and cerebrospinal fluid. Consequently, the gold standard for pre-surgical evaluation in drug-resistant focal epilepsy is stereo-electroencephalography (sEEG), an invasive procedure in which multi-contact depth electrodes (shafts) are stereotactically implanted directly into deep and superficial brain structures [1]. Because sEEG samples the local field potentials (LFPs) of neural populations at millisecond resolution right at the core of the epileptogenic network, it provides an ideal empirical testbed for applying dynamic graph theory to real-world biological phase transitions.

1.2 | From Time Series to Functional Networks

To translate the clinical phenomena of pre-ictal decoupling, modular breakdown, and ictal hypersynchronization into rigorous physical observables, we map the continuous multi-channel sEEG recordings onto the mathematical framework of Complex Network Theory [3, 9]. Rather than analyzing each intracranial electrode contact as an isolated time series, we treat the set of recorded brain sites as an interacting graph whose collective state is encoded in the time-varying topology of its functional links [4]. Every graph-theoretical metric introduced below is specifically selected to answer a distinct neurophysiological question across the macroscopic, mesoscopic, and microscopic scales of the seizure transition.

1.2.1 | Fundamentals of Graph Theory and Signed Dynamic Connectivity

Formally, a network is represented as a graph $G = (V, E)$, where $V = \{1, 2, \ldots, N\}$ is the set of $N = |V|$ vertices (or nodes) and $E \subseteq V \times V$ is the set of $L = |E|$ edges (or links) representing pairwise interactions between nodes [3]. In our neurophysiological context, each node $i \in V$ corresponds to a distinct intracranial sEEG recording contact, while an edge $(i, j) \in E$ represents a functional dependency between the electrical potentials recorded at contacts i and j.

While anatomical (structural) brain networks map physical axonal tracts via diffusion imaging, Functional Connectivity (FC) quantifies the statistical interdependence between physiological signals recorded across distributed sites [4]. Because the transition from the inter-ictal baseline to an epileptic seizure is fundamentally non-stationary, computing a single static connectivity matrix over the entire recording would average out the rapid bifurcations occurring around seizure onset. To resolve the temporal trajectory of the network, we adopt a Dynamic Functional Connectivity (dFC) approach: the continuous multi-channel voltage recordings $x_i(t)$ are segmented into a sequence of overlapping sliding windows of fixed duration T and step size $\Delta\tau$, centered at discrete time instants τ.

Within each temporal window $T_\tau = [\tau - T/2, \tau + T/2]$, the pairwise functional coupling between contact i and contact j is measured via the zero-lag sample Pearson correlation coefficient:

$$W_{ij}(\tau) = \frac{\sum_{t \in T_\tau} \left(x_i(t) - \bar{x}_i(\tau)\right)\left(x_j(t) - \bar{x}_j(\tau)\right)}{\sqrt{\sum_{t \in T_\tau} \left(x_i(t) - \bar{x}_i(\tau)\right)^2} \sqrt{\sum_{t \in T_\tau} \left(x_j(t) - \bar{x}_j(\tau)\right)^2}} \tag{1.2.1}$$

where $\bar{x}_i(\tau) = \frac{1}{|T_\tau|} \sum_{t \in T_\tau} x_i(t)$ is the local temporal mean of channel i inside window T_τ. Since Pearson correlation is invariant under positive linear transformations of the data, $W_{ij}(\tau)$ is bounded within $[-1, 1]$ and measures the degree of linear phase-locking between the two neural populations, normalized by their local signal variance. Self-interactions are excluded by imposing a zero diagonal, $W_{ii}(\tau) = 0$. Across all sliding windows, this procedure yields a three-dimensional, symmetric, signed dynamic connectivity tensor $\mathbf{W} \in [-1, 1]^{T_{\text{wins}} \times N \times N}$, where $W_{ij}(\tau) = W_{ji}(\tau)$.

A central methodological pillar of this study is the **strict preservation of the algebraic sign** of $W_{ij}(\tau)$. Standard functional connectivity pipelines frequently take the absolute value $|W_{ij}(\tau)|$ prior to graph construction, operating under the simplifying assumption that any high correlation magnitude reflects strong functional coupling. However, from both a physical and neurophysiological standpoint, positive and negative correlations correspond to fundamentally opposing dynamical regimes:

- **Cooperative In-Phase Synchronization ($W_{ij}(\tau) > 0$):** Reflects coherent, zero-lag oscillatory alignment between neural populations. This regime drives the pathological coalescence of cortical hubs into a macroscopic syncytium during ictal onset and propagation.
- **Competitive Counter-Phase Decoupling ($W_{ij}(\tau) < 0$):** Reflects anti-phase dynamics, phase opposition, or active functional detachment between brain regions. Retaining negative correlations is indispensable to test the clinical hypothesis of pre-ictal isolation [8, 7]: if an epileptogenic focus detaches from the surrounding baseline network by entering a counter-phase oscillatory regime prior to onset, taking $|W_{ij}(\tau)|$ would erroneously conflate this premonitory decoupling with cooperative synchronization.

Consequently, rather than folding the negative spectrum onto the positive axis, we explicitly decompose the signed tensor $W_{ij}(\tau)$ into cooperative ($W_{ij} > 0$) and competitive ($W_{ij} < 0$) interaction layers, analyzing their global competition and nodal trajectories separately.

1.2.2 | Macroscopic Scale: Percolation Theory and Critical Susceptibility

At the macroscopic scale, our objective is to determine whether the onset of a focal seizure behaves as a collective phase transition, quantifying how localized neural discharges break global symmetry to invade the entire cerebral network. In statistical mechanics and network science, the sudden emergence of macroscopic long-range connectivity across a system as link density increases is described by *percolation theory* [3, 9].

To cast our time-varying functional tensor into the framework of bond percolation, we map each continuous correlation matrix $W(\tau)$ into a sparse, unweighted binary graph $G(\tau) = (V, E(\tau))$ described by a symmetric binary Adjacency Matrix $A(\tau) \in \{0,1\}^{N\times N}$. Because ictal recruitment is driven by cooperative in-phase locking, we apply a strict positive correlation threshold $\theta \in (0,1)$ via the Heaviside step function $\Theta(\cdot)$:

$$A_{ij}(\tau) = \Theta(W_{ij}(\tau) - \theta) = \begin{cases} 1 & \text{if } W_{ij}(\tau) \geq \theta \text{ and } i \neq j \\ 0 & \text{otherwise} \end{cases} \tag{1.2.2}$$

Similarly, to monitor the global density of competitive counter-phase links, one can define a negative threshold $\theta_{\text{neg}} < 0$ and track the fraction of pairs satisfying $W_{ij}(\tau) \leq \theta_{\text{neg}}$. Within

the cooperative binary graph $A(\tau)$, the local connectivity of contact i is given by its node degree $k_i(\tau)$, which counts the number of active cooperative links incident to node i:

$$k_i(\tau) = \sum_{j=1}^{N} A_{ij}(\tau) \tag{1.2.3}$$

and the overall density of active cooperative links in the network at time τ is given by $D^+(\tau) = \frac{1}{N(N-1)} \sum_{i,j} A_{ij}(\tau)$.

In classical percolation theory, as the control parameter (either the threshold θ or the endogenous neural synchronization dictated by time τ) is varied, the network transitions between two topological regimes [3, 2]:

1. **Subcritical (Fragmented) Phase:** During the physiological inter-ictal baseline, strong correlations ($W_{ij} \geq \theta$) are sparse and confined to anatomically adjacent contacts. The graph decomposes into numerous small, mutually disconnected connected components (islands), and the largest component encompasses only a negligible fraction of the total nodes ($N_{\text{GCC}} \ll N$).

2. **Supercritical (Percolating) Phase:** As the seizure onset is crossed, widespread cortical areas lock in phase, causing previously isolated clusters to coalesce rapidly into a single macroscopic *Giant Connected Component* (GCC) that spans a finite, macroscopic fraction of the entire recording array ($N_{\text{GCC}} \sim \mathcal{O}(N)$).

In direct analogy with the spontaneous magnetization M in the ferromagnetic Ising model, the macroscopic **order parameter** of the percolation transition is the normalized size of the Giant Connected Component, $S(\tau)$:

$$S(\tau) = \frac{N_{\text{GCC}}(\tau)}{N} \in \left[\frac{1}{N}, 1\right] \tag{1.2.4}$$

where $N_{\text{GCC}}(\tau)$ is the number of nodes belonging to the largest connected subgraph of $G(\tau)$ at window τ.

Furthermore, a hallmark of critical phenomena in statistical mechanics is the divergence (or sharp finite-size peaking) of fluctuations in the vicinity of the critical point. When the cortical network is poised right at the onset bifurcation— transitioning from fragmented clusters to a giant syncytium—nodes exhibit maximal heterogeneity in their connectivity: highly recruited core hubs coexist with still-unrecruited peripheral contacts. We quantify these critical topological fluctuations via the **topological susceptibility** $\chi(\tau)$, defined as the spatial variance of the binary degree distribution across the N nodes at time τ:

$$\chi(\tau) = \text{Var}(k(\tau)) = \frac{1}{N} \sum_{i=1}^{N} \left(k_i(\tau) - \langle k(\tau) \rangle\right)^2, \quad \text{where } \langle k(\tau) \rangle = \frac{1}{N} \sum_{i=1}^{N} k_i(\tau) \tag{1.2.5}$$

Tracking $S(\tau)$ alongside $\chi(\tau)$ allows us to verify whether seizure onset manifests as a genuine critical percolation transition, marked by a simultaneous jump in the order parameter $S(\tau)$ and a sharp peak in degree fluctuations $\chi(\tau)$.

1.2.3 | Mesoscopic Scale: Community Structure and Modularity Collapse

While macroscopic percolation captures the global coalescence of the network into a giant component, it treats the graph as a homogeneous whole and remains blind to intermediate organizational scales. Crucially, healthy brain networks exhibit a pronounced *mesoscopic* architecture: cortical regions organize into functional modules (or *communities*), characterized by dense internal connections among nodes within the same functional subsystem and sparse connections between nodes belonging to different subsystems [4, 3]. This modular segregation prevents local perturbations from cascading uncontrollably across the entire cortex.

From a network dynamics standpoint, we hypothesize that the transition to an epileptic seizure entails a catastrophic breakdown of this mesoscopic segregation: as pathological synchronization spreads, distinct functional communities dissolve and merge into an undifferentiated global cluster. To test this quantitatively without arbitrary binary thresholding, we analyze the positive weighted cooperative network $W_{ij}^{+}(\tau) = \max(0, W_{ij}(\tau))$.

Let $\mathcal{C}(\tau) = \{c_1(\tau), c_2(\tau), \ldots, c_N(\tau)\}$ denote a partition of the N nodes into disjoint communities at time window τ, where $c_i(\tau)$ identifies the module to which contact i is assigned. The quality of this partition—and thus the degree of functional segregation of the underlying brain state—is measured by **Newman's weighted Modularity** $Q(\tau)$ [3]:

$$Q(\tau) = \frac{1}{2m(\tau)} \sum_{i=1}^{N} \sum_{j=1}^{N} \left[W_{ij}^{+}(\tau) - \frac{s_i(\tau) s_j(\tau)}{2m(\tau)} \right] \delta(c_i(\tau), c_j(\tau)) \tag{1.2.6}$$

where:

- $s_i(\tau) = \sum_{j=1}^{N} W_{ij}^{+}(\tau)$ is the positive weighted degree (*node strength*) of contact i;
- $m(\tau) = \frac{1}{2} \sum_{i,j=1}^{N} W_{ij}^{+}(\tau) = \frac{1}{2} \sum_{i=1}^{N} s_i(\tau)$ is the total weight of all cooperative links in the network;
- $\delta(c_i, c_j)$ is the Kronecker delta function, equal to 1 if nodes i and j belong to the same community ($c_i = c_j$) and 0 otherwise.

The physical interpretation of Equation (1.2.6) lies in the comparison against a statistical null model. The term $P_{ij}(\tau) = \frac{s_i(\tau) s_j(\tau)}{2m(\tau)}$ represents the expected edge weight between node i and node j under the *configuration model*—a randomized null network that preserves the exact strength sequence $\{s_1, \ldots, s_N\}$ of the empirical graph but rewires connections uniformly at random [3, 9]. Thus, $Q(\tau) \in [-1/2, 1]$ measures the excess fraction of intracommunity weight relative to random chance:

- **High Modularity ($Q(\tau) \gtrsim 0.3$):** Indicates a strongly segregated mesoscopic architecture in which functional interactions are tightly confined within specialized local modules (typical of the physiological baseline).

- **Modularity Collapse ($Q(\tau) \to 0$):** Indicates that internal community boundaries have dissolved and connectivity has become globally homogeneous, as expected when the ictal wave merges previously independent subsystems.

Because finding the exact partition $\mathcal{C}(\tau)$ that maximizes $Q(\tau)$ over all possible node assignments is an NP-hard combinatorial optimization problem, we employ the **Louvain heuristic algorithm** [10]. The Louvain method operates hierarchically by alternating two rapid phases:

1. **Local Modularity Optimization:** Starting with each node in its own singleton community (N communities), the algorithm iterates sequentially over all nodes $i \in V$, evaluating the modularity gain ΔQ obtained by moving node i from its current community into the community of each of its neighbors j. Node i is reassigned to the neighbor's community that yields the largest positive gain $\Delta Q > 0$. This sweep is repeated until a local maximum of Q is reached and no single-node move can improve the partition.

2. **Coarse-Graining (Network Aggregation):** The communities identified in the first phase are collapsed into "super-nodes" to build a new, reduced weighted network. The edge weight between two super-nodes is given by the sum of all link weights between nodes in the corresponding communities, while intracommunity links become self-loops on the super-nodes. Phase 1 is then re-applied to this coarse-grained graph until modularity converges.

Tracking both $Q(\tau)$ and the number of detected communities $N_c(\tau)$ across the transition provides a direct mesoscopic readout of functional unbinding and ictal coalescence.

1.2.4 | Microscopic Scale: Eigenvector Centrality and Premonitory Decoupling

While $S(\tau)$ and $Q(\tau)$ characterize the global and modular state of the brain, clinical epileptology ultimately requires localizing the specific recording contacts that drive the seizure. At the microscopic (nodal) level, however, a single topological metric is insufficient because channels play distinct dynamical roles before and during the phase transition [11]. Specifically, we must mathematically distinguish between: (i) *premonitory decoupling nodes*, which isolate themselves in counter-phase during the pre-ictal window, and (ii) *ictal hub nodes*, which dominate the cooperative Giant Connected Component once the seizure erupts.

To identify the dominant cooperative hubs during seizure onset, simple binary degree $k_i(\tau)$ or strength $s_i(\tau)$ is a naive estimator: a node with moderate degree connected to the core drivers of the seizure exercises far greater dynamical influence over network synchronization than a node connected to an equal number of isolated peripheral contacts. To capture

this recursive topological importance, we compute the **Leading Eigenvector Centrality** $\mathbf{v}_1(\tau)$ of the cooperative adjacency matrix $A(\tau)$ [3, 11].

In eigenvector centrality, the centrality score $v_{1,i}(\tau)$ of node i is defined as being proportional to the sum of the scores of its connected neighbors:

$$v_{1,i}(\tau) = \frac{1}{\lambda_1(\tau)} \sum_{j=1}^{N} A_{ij}(\tau)\, v_{1,j}(\tau) \quad \Longleftrightarrow \quad \mathbf{A}(\tau)\mathbf{v}_1(\tau) = \lambda_1(\tau)\mathbf{v}_1(\tau) \tag{1.2.7}$$

where $\lambda_1(\tau)$ is the largest (leading) real eigenvalue of the symmetric matrix $\mathbf{A}(\tau)$. By the *Perron-Frobenius theorem* for non-negative matrices, the leading eigenvector $\mathbf{v}_1(\tau)$ associated with $\lambda_1(\tau)$ possesses strictly non-negative components ($v_{1,i}(\tau) \geq 0$) and uniquely concentrates its weight on the most densely interconnected core of the Giant Connected Component [3]. To ensure that centrality scores represent relative nodal dominance and remain directly comparable across time windows with vastly different total link densities, we normalize $\mathbf{v}_1(\tau)$ at every step using the L_1-norm:

$$\sum_{i=1}^{N} v_{1,i}(\tau) = 1 \tag{1.2.8}$$

Conversely, because eigenvector centrality operates on the positive cooperative graph $A(\tau)$, it cannot see the competitive counter-phase detachment occurring prior to onset. To quantify the clinical hypothesis of pre-ictal functional isolation [8, 7], we turn to the negative domain of our signed connectivity tensor $W(\tau)$. For each node i and time window τ, we define its instantaneous **Negative Decoupling Score** $d_i^{(-)}(\tau)$ as the mean magnitude of its negative (counter-phase) correlations with all other $N-1$ nodes in the network:

$$d_i^{(-)}(\tau) = \frac{1}{N-1} \sum_{j \neq i} |\min(0,\, W_{ij}(\tau))| \in [0, 1] \tag{1.2.9}$$

A high value of $d_i^{(-)}(\tau)$ during the pre-ictal window indicates that contact i is actively oscillating against the global cortical background, detaching itself from physiological homeostasis.

Finally, because instantaneous network metrics computed over short sliding windows ($T \sim 2$ s) are inevitably subject to stochastic physiological fluctuations, evaluating node roles at arbitrary single time points (τ) would be vulnerable to transient noise spikes. To achieve robust statistical inference, we integrate these microscopic observables over well-defined macro-epochs $\mathcal{E}$ (specifically a quiet pre-ictal *Baseline* epoch $\mathcal{E}_{\text{base}}$, a *Premonitory* window $\mathcal{E}_{\text{prem}}$ immediately preceding $t = 0$, and an *Onset* epoch $\mathcal{E}_{\text{onset}}$ immediately following $t = 0$). By pairing each channel's epoch-averaged premonitory decoupling score:

$$\langle d_i^{(-)} \rangle_{\text{prem}} = \frac{1}{|\mathcal{E}_{\text{prem}}|} \sum_{\tau \in \mathcal{E}_{\text{prem}}} d_i^{(-)}(\tau) \tag{1.2.10}$$

with its epoch-averaged onset centrality gain over baseline:

$$\Delta \bar{v}_{1,i} = \langle v_{1,i} \rangle_{\text{onset}} - \langle v_{1,i} \rangle_{\text{base}} \tag{1.2.11}$$

we establish a noise-resilient, two-dimensional feature space—the **Microscopic Functional Role Map**—capable of disentangling premonitory seizure initiators from secondary cortical amplifiers.

2 | Materials, Methods, and Computational Implementation

Translating the theoretical framework of Chapter 1 into a reproducible empirical analysis requires addressing three practical challenges inherent to clinical stereo-electroencephalography (sEEG): (i) the computational overhead of handling high-frequency, multi-channel recordings of several gigabytes; (ii) the contamination of intracranial potentials by extracranial polygraphic artifacts; and (iii) the non-stationarity and inter-patient variability of clinical metadata. This chapter details the dataset structure, the signal preprocessing pipeline, and the core numerical algorithms implemented in Python to extract and calibrate the dynamic functional networks.

2.1 | The sEEG Dataset and Memory-Mapped Extraction

The empirical dataset analyzed in this study consists of invasive multi-channel sEEG recordings acquired from patients with drug-resistant focal epilepsy undergoing pre-surgical evaluation. Each recording is stored as a compressed NumPy archive (`.npz`) containing the continuous voltage matrix $\mathbf{X} \in \mathbb{R}^{N_{\text{raw}} \times M_{\text{tot}}}$, sampled at $f_s = 1000$ Hz, alongside an array of clinical channel labels and a metadata table specifying the annotated seizure onset time t_{onset}. The number of implanted intracranial contacts per patient ranges between $N \approx 110$ and 170, distributed across multiple stereotactic depth electrode shafts (typically denoted by alphabetical prefixes such as `A`, `B`, `H`, `S'`, `Y'`, where prime symbols indicate left-hemisphere implants).

2.1.1 | Out-of-Core Slicing via Memory Mapping

Because continuous clinical monitoring spans hours of activity at kilohertz resolution, loading an entire raw recording array directly into volatile memory (RAM) is computationally inefficient and prone to memory overflow. However, the critical dynamical reorganization governing the transition from the inter-ictal baseline to the ictal state unfolds over a temporal horizon of a few minutes around seizure onset.

To isolate the phase transition with minimal memory footprint, we access the raw archives in read-only memory-mapped mode (`mmap_mode="r"`). This operating system mechanism maps the byte offsets of the uncompressed array directly onto virtual address space without transferring the full file into RAM. Given the aligned seizure onset timestamp t_{onset}, we slice a symmetric 240 s transition window spanning $\Delta t_{\text{pre}} = 120$ s prior to onset and $\Delta t_{\text{post}} = 120$ s following onset, shifting the temporal axis so that $t = 0$ s corresponds exactly to the transition point.

2.1.2 | Artifact Channel Filtering and Channel-Wise Standardization

A crucial preprocessing step involves cleaning the node set V from non-cerebral recording channels. Clinical sEEG systems routinely record auxiliary polygraphic channels alongside intracranial depth electrodes, labeled with prefixes such as `POL` (polygraphy/polarity references), `ECG` (electrocardiogram), or `MK` (event markers). During the pre-ictal and ictal phases, autonomic changes (e.g., ictal tachycardia) and subtle patient motor activity generate high-amplitude, low-frequency electrical fluctuations on these extracranial channels. If retained in the graph, such signals exhibit strong spurious anticorrelations with cortical activity, falsely appearing as decoupled network hubs. We therefore programmatically filter out all non-intracranial labels prior to network construction.

Subsequently, because electrode impedance and local tissue conductivity vary across recording sites, raw voltage amplitudes cannot be compared directly without normalization.

Each retained intracranial channel $i \in \{1, \ldots, N\}$ is independently standardized over the extracted 240 s chunk via a channel-wise z-score transformation:

$$z_i(t) = \frac{x_i(t) - \mu_i}{\sigma_i} \tag{2.1.1}$$

where μ_i and σ_i denote the temporal mean and standard deviation of channel i across the extracted transition window. The core Python implementation of this memory-mapped extraction, channel filtering, and standardization procedure is shown below:

```python
def extract_transition_chunk(file_path, onset_sec, pre_sec=120.0, post_sec=120.0):
    # Access large archive on disk without loading full array into RAM
    data = np.load(file_path, mmap_mode="r")
    sfreq = float(data["sfreq"]) if "sfreq" in data.files else 1000.0
    signal_arr = data[data.files[0]]
    channel_names = data["channels"] if "channels" in data.files else np.array([])

    if signal_arr.shape[0] > signal_arr.shape[1]:
        signal_arr = signal_arr.T  # Enforce (N_channels, N_samples)

    start_idx = max(0, int((onset_sec - pre_sec) * sfreq))
    end_idx = min(signal_arr.shape[1], int((onset_sec + post_sec) * sfreq))

    # Pull only the target [-120s, +120s] window into memory
    raw_chunk = np.array(signal_arr[:, start_idx:end_idx], dtype=np.float64)

    # Filter out extracranial/polygraphic channels (POL, ECG, MK)
    if len(channel_names) > 0:
        valid_idx = [
            i for i, ch in enumerate(channel_names)
            if not any(tag in str(ch).upper() for tag in ("POL", "ECG", "MK"))
        ]
        raw_chunk = raw_chunk[valid_idx, :]
        channel_names = channel_names[valid_idx]

    # Channel-wise z-score standardization
    means = np.mean(raw_chunk, axis=1, keepdims=True)
    stds = np.std(raw_chunk, axis=1, keepdims=True)
    stds[stds == 0.0] = 1.0
    normalized_chunk = (raw_chunk - means) / stds

    rel_time = (np.arange(start_idx, end_idx) - int(onset_sec * sfreq)) / sfreq
    return normalized_chunk, rel_time, sfreq, channel_names
```

Listing 2.1: Memory-mapped transition window extraction, extracranial channel filtering, and channel-wise z-score normalization.

2.2 | Sliding-Window Tensor Construction

To track the non-stationary topological evolution of the sEEG network across the 240 s window, the standardized signal matrix $\mathbf{Z} \in \mathbb{R}^{N \times M}$ is partitioned into overlapping sliding windows of fixed duration $T = 2.0$ s (2000 samples) and step size $\Delta\tau = 0.5$ s (500 samples). This parameterization yields a total of $T_{\text{wins}} = 477$ temporal frames centered at $\tau \in [-119.0 \text{ s}, +119.0 \text{ s}]$.

The choice of $T = 2.0$ s represents an optimal physical trade-off between statistical stability and temporal resolution: a 2000-sample window is sufficiently long to provide robust estimation of zero-lag Pearson correlations across physiological frequency bands (from delta to high-gamma), while a 0.5 s stride resolves rapid pre-ictal decoupling and onset recruitment dynamics.Crucially, both T and $\Delta\tau$ are held strictly constant across all patients to ensure that the variance of the sample correlation estimator remains invariant across the cohort.

Within each window T_τ, we compute the signed Pearson correlation matrix $\mathbf{W}(\tau) \in [-1, 1]^{N \times N}$ according to Equation (1.2.1), setting the diagonal elements to zero ($W_{ii}(\tau) = 0$) and clipping numerical floating-point inaccuracies to $[-1.0, 1.0]$. As established in Section 1.2, no absolute value transformation is applied, preserving the algebraic distinction between cooperative in-phase links ($W_{ij} > 0$) and competitive counter-phase links ($W_{ij} < 0$).

```python
def compute_dynamic_functional_connectivity(signal, sfreq, window_sec=2.0, step_sec=0.5):
    n_nodes, n_samples = signal.shape
    win_samples = int(window_sec * sfreq)
    step_samples = int(step_sec * sfreq)
    starts = np.arange(0, n_samples - win_samples + 1, step_samples)

    W = np.zeros((len(starts), n_nodes, n_nodes), dtype=np.float64)
    centers_sec = (starts + win_samples / 2.0) / sfreq

    for idx, s in enumerate(starts):
        segment = signal[:, s : s + win_samples]
        corr = np.corrcoef(segment)
        corr = np.nan_to_num(corr, nan=0.0)
        np.fill_diagonal(corr, 0.0)
        W[idx] = np.clip(corr, -1.0, 1.0)

    return W, centers_sec
```

Listing 2.2: Sliding-window computation of the signed dynamic functional connectivity tensor $\mathbf{W}(\tau)$.

2.3 | Data-Driven Calibration and Statistical Epochs

Applying network analysis to clinical recordings across multiple subjects introduces three fundamental methodological challenges:

(i) clinical onset timestamps annotated in hospital records often reflect the appearance of behavioral symptoms (*clinical onset*) rather than the deep intracranial onset of synchronous discharges (*electrophysiological onset*);

(ii) baseline functional connectivity density varies across patients due to individual differences in electrode implantation geometry;

(iii) evaluating nodal metrics at single time frames is highly sensitive to transient inter-ictal spikes.

We address each of these factors through objective, data-driven calibration criteria.

2.3.1 | Electrophysiological Onset Alignment

Upon inspecting the raw signals and preliminary connectivity traces across the dataset, notable discrepancies emerged between the nominal metadata timestamps and the actual dynamical bifurcation in specific recordings (where the macroscopic surge in synchronization occurred several seconds away from the nominal marker). Because evaluating pre-ictal metrics in a window $[-15, 0]$ s requires $t = 0$ s to mark the exact point of symmetry breaking, relying on delayed clinical annotations would erroneously contaminate the premonitory window with early ictal discharges.

To establish an objective, operator-independent definition of $t = 0$ s, we implemented an automated transition detector that scans the recording for the point of maximum upward gradient in both signal energy and network synchronization. Specifically, at each sliding window τ, we compute:

1. **Core Signal Energy $E_{\mathbf{RMS}}(\tau)$:** The root-mean-square (RMS) voltage amplitude averaged across the top 15% most active channels in window τ, capturing the sudden physical surge in local field potential variance generated by hypersynchronous neuronal firing in the Seizure Onset Zone.

2. **Global Cooperative Density $D^{+}_{0.5}(\tau)$:** The fraction of node pairs exhibiting strong in-phase correlation $W_{ij}(\tau) \geq 0.50$.

Both profiles are min-max normalized to $[0, 1]$ and averaged into a composite transition signal $\Phi(\tau) = \frac{1}{2}\hat{E}_{\mathrm{RMS}}(\tau) + \frac{1}{2}\hat{D}^{+}_{0.5}(\tau)$. The true electrophysiological onset τ^* is then identified as the argument maximizing the forward-backward step derivative over a $\Delta H = 6.0$ s

horizon:

$$\tau^* = \arg\max_{\tau} \left[\frac{1}{\Delta H} \int_{\tau}^{\tau+\Delta H} \Phi(t)\, dt - \frac{1}{\Delta H} \int_{\tau-\Delta H}^{\tau} \Phi(t)\, dt \right] \tag{2.3.1}$$

Integrating over a 6.0 s horizon acts as a low-pass filter on the derivative: brief, isolated inter-ictal spikes (lasting < 1 s) cancel out, whereas the sustained step transition into the ictal state produces a sharp global maximum, aligning $t = 0$ s accurately across all recordings.

2.3.2 | Sub-Critical Threshold Calibration and Statistical Epochs

In percolation analysis, applying an arbitrary universal threshold θ across patients with different implantation densities can distort topological comparisons: a recording with densely spaced contacts may already exhibit a percolating giant component during resting state at $\theta = 0.50$, whereas a sparse implant may remain overly fragmented. Following percolation theory, we calibrate $\theta \in \{0.45, 0.50, 0.55, 0.60, 0.65\}$ for each subject using an objective baseline criterion: θ is selected so that the cooperative network during the quiet inter-ictal baseline resides right at the boundary of the sub-critical regime, defined by a mean positive edge density of $D^{+}_{\text{base}}(\theta) \approx 6\%$–$8\%$.

Finally, rather than comparing node centralities at arbitrary single snapshots, we partition the aligned temporal axis $\tau \in [-120\text{ s}, +120\text{ s}]$ into four functionally distinct statistical epochs:

1. **Quiet Baseline Epoch ($\mathcal{E}_{\text{base}}$, typically $[-90, -20]$ s):** A 50–70 s reference window selected prior to $\tau = -20$ s by minimizing a cost function of mean synchronization and RMS variance, thereby automatically avoiding isolated pre-ictal bursts.

2. **Premonitory Window ($\mathcal{E}_{\text{prem}} = [-15.0, 0.0]$ s):** The immediate pre-onset interval where active counter-phase decoupling $\langle d_i^{(-)} \rangle_{\text{prem}}$ is evaluated to test for pre-ictal isolation of the epileptogenic focus.

3. **Onset Recruitment Epoch ($\mathcal{E}_{\text{onset}} = [0.0, +15.0]$ s):** The early ictal bifurcation window over which leading eigenvector centrality $\langle v_{1,i} \rangle_{\text{onset}}$ is averaged to identify the primary hub drivers of the Giant Connected Component.

4. **Established Ictal Epoch ($\mathcal{E}_{\text{ictal}} = [+15.0, +60.0]$ s):** The fully developed seizure propagation phase.

Computing microscopic node scores as statistical expectations over these epochs suppresses stochastic frame-to-frame fluctuations and guarantees robust role identification.

3 | Multiscale Analysis of the Ictal Transition

Having established the theoretical observables in Chapter 1 and the data-driven preprocessing pipeline in Chapter 2, we now present the empirical investigation of the pre-ictal to ictal transition. We structure this chapter as a progressive multiscale zoom—moving from the macroscopic global graph, through mesoscopic community organization, down to microscopic node trajectories and functional role mapping. At each scale, we pair the core numerical implementation with the resulting dynamical profiles, concluding with a critical assessment of the methodological limitations that motivate the transition to the next scale of resolution.

Before decomposing the transition into specific topological metrics, it is instructive to inspect the raw structure of the signed dynamic functional connectivity tensor $\mathbf{W}(\tau) \in [-1, 1]^{N \times N}$ directly. Figure 3.1 provides a panoramic overview of the connectivity matrices at three milestone time instants—Pre-Ictal ($t = -60.0$ s), Seizure Onset ($t = 0.0$ s), and Established Ictal ($t = +20.0$ s)—alongside the continuous temporal evolution of active cooperative links ($W_{ij} \geq 0.55$) and competitive anticorrelated links ($W_{ij} \leq -0.35$).

The three matrix snapshots in the top row of Figure 3.1 visually justify the core methodological premises of this work:

- **Pre-Ictal Baseline ($t = -60.0$ s):** The correlation matrix is dominated by pale, near-zero background values ($W_{ij} \approx 0$), with strong positive correlations restricted to small, block-diagonal squares along the main diagonal. These local blocks correspond to contacts situated along the same stereotactic depth electrode shaft, reflecting physiological local segregation and spatial volume conduction.
- **Seizure Onset Bifurcation ($t = 0.0$ s):** Exactly at the aligned transition point, off-diagonal structure erupts across the matrix. Crucially, the matrix does not turn uniformly red: instead, a striking checkerboard pattern of intense red blocks ($W_{ij} > +0.75$) and deep blue stripes ($W_{ij} < -0.60$) emerges. This confirms that as the ictal core locks into in-phase synchronization, specific subsets of channels simultaneously enter strong counter-phase opposition against the rest of the cortex. Had we applied the conventional absolute value rectification $|W_{ij}|$, these blue counter-

phase stripes would have been folded into positive weights, completely obscuring the competitive geometry of the transition.

- **Established Ictal State ($t = +20.0$ s) and Global Density Evolution:** By $t = +20.0$ s, massive red off-diagonal blocks span almost the entire matrix, interrupted only by a few persistent blue counter-phase bands (around channel indices 35–45). Consistently, the bottom panel shows that both cooperative links (blue curve) and anticorrelated links (red curve) remain low and stable ($\sim$ 3–10%) throughout the entire pre-ictal period, before jumping sharply at $t = 0$ s into a sustained, high-density regime (20–42%) that persists throughout the seizure.

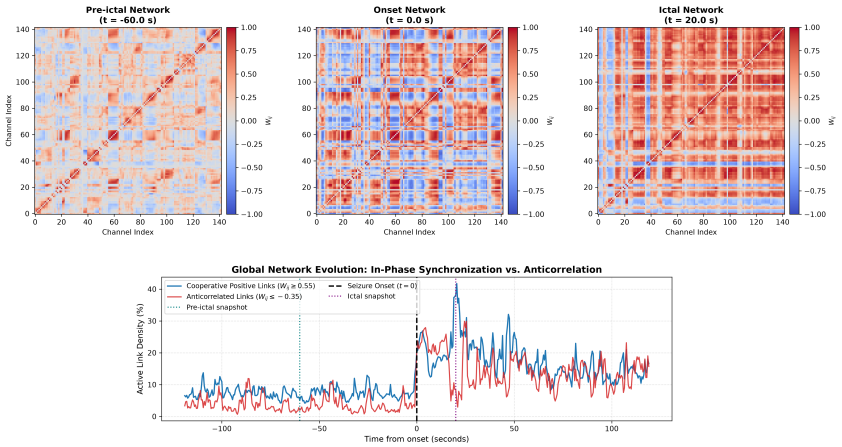


Figure 3.1: Panoramic overview of the signed dynamic functional connectivity tensor $\mathbf{W}(\tau)$. Top row: Full $N \times N$ correlation matrices ($N \approx 140$ intracranial channels) sampled during the pre-ictal baseline ($t = -60.0$ s), seizure onset ($t = 0.0$ s), and established ictal propagation ($t = +20.0$ s). Warm colors ($W_{ij} > 0$, red) denote in-phase cooperative synchronization; cool colors ($W_{ij} < 0$, blue) denote counter-phase anticorrelation. Bottom panel: Global temporal evolution of the active cooperative link density ($W_{ij} \geq 0.55$, blue trace) versus competitive anticorrelated link density ($W_{ij} \leq -0.35$, red trace) across the 240 s window.

With this global intuition established, we now proceed to quantify the transition systematically across the three topological scales on our reference recording (`DB470450`), followed by a cross-patient robustness analysis in Section 3.4.

3.1 | Macroscopic Scale: Percolation Phase Transition and Critical Susceptibility

3.1.1 | Physical Motivation

At the coarsest level of resolution, our primary question is whether the onset of a focal seizure manifests macroscopically as a collective phase transition. During the physiological inter-ictal baseline, functional coupling between distinct sEEG contacts is expected to remain weak, transient, and spatially localized, keeping the global graph in a fragmented, subcritical regime. As the epileptogenic zone overcomes local inhibitory restraint at $t = 0$ s, we test whether the cortical network undergoes a percolation transition characterized by: (i) the sudden macroscopic emergence of a Giant Connected Component $S(\tau)$, (ii) a sharp divergence in topological susceptibility $\chi(\tau) = \mathrm{Var}(k(\tau))$ marking critical degree heterogeneity, and (iii) a global divergence between cooperative in-phase synchronization and competitive counter-phase decoupling.

3.1.2 | Numerical Implementation

To evaluate percolation metrics across the $T_{\text{wins}} = 477$ sliding windows without introducing spatial priors, each frame $\mathbf{W}(\tau)$ of the signed connectivity tensor is thresholded at the calibrated critical value θ ($\theta = 0.50$ for `DB470450`). The size of the Giant Connected Component $N_{\text{GCC}}(\tau)$ is extracted via breadth-first graph traversal on the sparse binary adjacency matrix $\mathbf{A}(\tau)$, while the susceptibility $\chi(\tau)$ and the positive/negative edge densities ($D^+(\tau)$ for $W_{ij} \geq \theta$ and $D^-(\tau)$ for $W_{ij} \leq \theta_{\text{neg}} = -0.35$) are computed directly from the upper triangular entries of $\mathbf{W}(\tau)$:

```python
from scipy.sparse import csr_matrix
from scipy.sparse.csgraph import connected_components

def compute_macroscopic_percolation(W, pos_thresh=0.50, neg_thresh=-0.35):
    n_wins, n_nodes, _ = W.shape
    triu_idx = np.triu_indices(n_nodes, k=1)
    n_pairs = len(triu_idx[0])

    gcc_fraction = np.zeros(n_wins, dtype=np.float64)
    susceptibility = np.zeros(n_wins, dtype=np.float64)
    pos_density = np.zeros(n_wins, dtype=np.float64)
    neg_density = np.zeros(n_wins, dtype=np.float64)

    for t in range(n_wins):
        W_t = W[t]
        upper_vals = W_t[triu_idx]

```

```
        # Global cooperative vs. competitive edge densities
        pos_density[t] = np.sum(upper_vals >= pos_thresh) / n_pairs
        neg_density[t] = np.sum(upper_vals <= neg_thresh) / n_pairs

        # Binary cooperative adjacency matrix A(t)
        adj = (W_t >= pos_thresh).astype(np.int8)
        np.fill_diagonal(adj, 0)

        # Connected components via sparse breadth-first search
        n_comp, labels = connected_components(csr_matrix(adj), directed
    =False)
        _, counts = np.unique(labels, return_counts=True)
        gcc_fraction[t] = np.max(counts) / float(n_nodes)

        # Topological susceptibility as spatial degree variance Var(k)
        degrees = np.sum(adj, axis=1)
        susceptibility[t] = np.var(degrees)

    return gcc_fraction, susceptibility, pos_density, neg_density
```

Listing 3.1: Macroscopic percolation metrics: Giant Connected Component $S(\tau)$, topological susceptibility $\chi(\tau)$, and signed edge densities.

3.1.3 | Empirical Results: Multi-Threshold Percolation and Susceptibility

Figure 3.2 illustrates the macroscopic percolation transition across the $[-120\text{ s}, +120\text{ s}]$ window by comparing the evolution of the Giant Connected Component $S(\tau)$ (top panel) and the topological susceptibility $\chi(\tau) = \mathrm{Var}(k(\tau))$ (bottom panel) across three distinct binarization thresholds: $\theta \in \{0.40, 0.50, 0.65\}$.

Comparing the three curves in the top panel of Figure 3.2 provides a textbook demonstration of how the choice of the control parameter θ determines the observability of the percolation phase transition:

- **Supercritical Saturation at Low Thresholds ($\theta = 0.40$ and $\theta = 0.50$):** For permissive thresholds (blue and orange traces), background physiological correlations and volume conduction are sufficient to maintain a spanning tree across the electrode array even during the inter-ictal baseline. Consequently, $S(\tau)$ remains pinned near 0.95–1.0 for $\theta = 0.40$ and fluctuates high between 0.80 and 0.98 for $\theta = 0.50$, largely masking the macroscopic jump at seizure onset.
- **Critical Percolation Regime ($\theta = 0.65$):** Raising the threshold to a stringent level ($\theta = 0.65$, purple trace) prunes weak background links and places the pre-ictal network genuinely within the *subcritical fragmented phase*, where $S(\tau)$ fluctuates broadly between 0.20 and 0.65. Exactly at the aligned electrophysiological onset ($t = 0$ s), the network undergoes a sharp discontinuous transition: $S(\tau)$ jumps to ~ 0.95–1.0

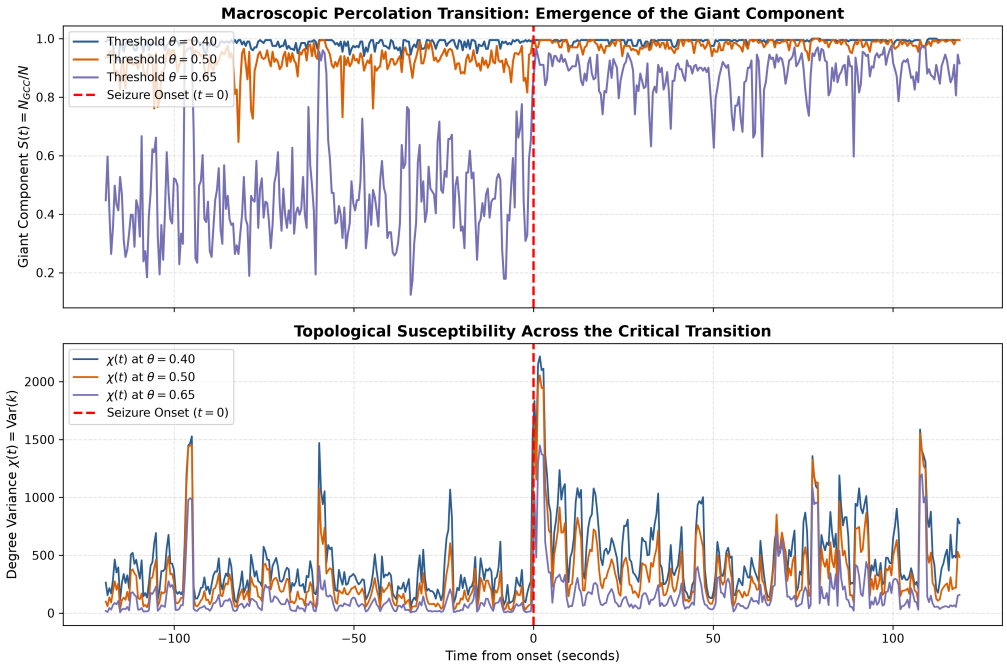


Figure 3.2: Macroscopic percolation transition across correlation thresholds $\theta \in \{0.40, 0.50, 0.65\}$. Top: Normalized Giant Connected Component size $S(\tau) = N_{\text{GCC}}/N$. Bottom: Topological susceptibility $\chi(\tau) = \text{Var}(k(\tau))$ peaking sharply at the aligned seizure onset ($t = 0$ s, red dashed line).

and remains stably in the percolating regime ($S(\tau) \in [0.80, 0.98]$) throughout the subsequent ictal propagation.

Remarkably, the bottom panel of Figure 3.2 reveals that while $S(\tau)$ is prone to ceiling saturation at lower thresholds, the topological susceptibility $\chi(\tau) = \text{Var}(k(\tau))$ acts as a threshold-robust indicator of criticality. During the quiet baseline, $\chi(\tau)$ remains low ($\chi < 400$ across all thresholds), interrupted only by two brief, isolated inter-ictal bursts at $\tau \approx -95$ s ($\chi \approx 1500$) and $\tau \approx -59$ s ($\chi \approx 1450$), where transient synchronization spikes and rapidly relaxes due to intact cortical inhibition. Exactly at seizure onset ($t \in [0, +3]$ s), however, $\chi(\tau)$ exhibits its absolute global maximum across all thresholds simultaneously—surging to $\chi \approx 2250$ for $\theta = 0.40$, $\chi \approx 2050$ for $\theta = 0.50$, and $\chi \approx 1450$ for $\theta = 0.65$. Physically, this sharp divergence in degree variance captures the critical spatial heterogeneity at the bifurcation point, where a hyper-connected core of ictal hub channels coexists with still-unrecruited peripheral contacts before the seizure settles into a fluctuating ictal plateau.

3.1.4 | Critical Assessment and Limitations of the Macroscopic View

While $S(\tau)$ and $\chi(\tau)$ confirm that seizure onset obeys the phenomenology of a critical percolation transition, relying exclusively on macroscopic binary descriptors presents two

significant limitations:

1. **Threshold Sensitivity:** Both $S(\tau)$ and $\chi(\tau)$ depend on the choice of the binarization threshold θ. If θ is set too low (e.g., $\theta < 0.35$), weak background correlations already percolate into a giant component during the inter-ictal state ($S(\tau) \approx 1$ everywhere), masking the onset jump. Conversely, if θ is set too high ($\theta > 0.75$), the graph remains fragmented even during the seizure. Although our data-driven calibration (targeting a subcritical baseline density of $\sim 7\%$) mitigates this arbitrariness, a threshold-free validation of network coalescence is desirable.

2. **Topological Blindness to Internal Structure:** Once $S(\tau)$ saturates near 1, percolation theory only tells us that a path exists between almost all nodes. It cannot distinguish whether the graph consists of several loosely connected functional modules (connected by a few bridge edges) or a genuinely collapsed, homogeneous syncytium.

To overcome threshold dependence and resolve the internal reorganization of functional subsystems, we must move to the mesoscopic scale.

3.2 | Mesoscopic Scale: Community Reorganization and Modularity Collapse

3.2.1 | Physical Motivation

As demonstrated in the macroscopic analysis, observing $S(\tau) \to 1$ at lower thresholds ($\theta \leq 0.55$) indicates that a path connects almost all nodes in the graph, yet it leaves open a fundamental topological question: does the pre-ictal network consist of distinct, weakly coupled functional modules that only merge into a single dense syncytium at $t = 0$ s?

To resolve this internal architecture, we move to the mesoscopic scale. In a healthy physiological state, cortical computation relies on functional segregation, which manifests graph-theoretically as a high Newman Modularity $Q(\tau)$ and a large number of autonomous communities $N_c(\tau)$. We hypothesize that the onset of the epileptic seizure induces a mesoscopic collapse, i.e. a sudden drop in $Q(\tau)$ accompanied by the coalescence of multiple pre-ictal modules into a minimal number of macro-communities dominated by the pathological Giant Component.

3.2.2 | Numerical Implementation

For each sliding window τ, we construct the thresholded positive weighted graph $W_{ij}^{(\theta)}(\tau) = W_{ij}(\tau)\,\Theta(W_{ij}(\tau) - \theta)$ at the calibrated mesoscopic threshold $\theta = 0.55$. Retaining the continuous weights $W_{ij}(\tau) \geq \theta$ preserves the relative interaction strengths within the active backbone while suppressing background correlation noise. We then apply the Louvain heuristic algorithm to extract the optimal partition $\mathcal{C}(\tau)$, recording both the maximum modularity score $Q(\tau)$ according to Equation (1.2.6) and the total number of detected communities $N_c(\tau) = |\mathcal{C}(\tau)|$:

```python
import networkx as nx
from networkx.algorithms.community import louvain_communities,
    modularity

def compute_mesoscopic_modularity(W, threshold=0.55, seed=42):
    n_wins, n_nodes, _ = W.shape
    modularity_q = np.zeros(n_wins, dtype=np.float64)
    n_communities = np.zeros(n_wins, dtype=np.int32)

    for t in range(n_wins):
        # Thresholded positive weighted adjacency matrix
        W_pos = np.where(W[t] >= threshold, W[t], 0.0)
        np.fill_diagonal(W_pos, 0.0)

        G = nx.from_numpy_array(W_pos)
        if G.size(weight="weight") == 0:
            n_communities[t] = n_nodes
```

```
            modularity_q[t] = 0.0
            continue

        # Louvain community partitioning and Newman's Q evaluation
        communities = louvain_communities(G, weight="weight", seed=seed
    )
        modularity_q[t] = modularity(G, communities, weight="weight")
        n_communities[t] = len(communities)

    return modularity_q, n_communities
```

Listing 3.2: Mesoscopic community detection via the Louvain algorithm and Newman's weighted modularity $Q(\tau)$.

3.2.3 | Empirical Results: Breakdown of Functional Segregation

Figure 3.3 displays the temporal evolution of Newman's Modularity $Q(\tau)$ (top panel) and the total number of detected communities $N_c(\tau)$ (bottom panel) at $\theta = 0.55$.

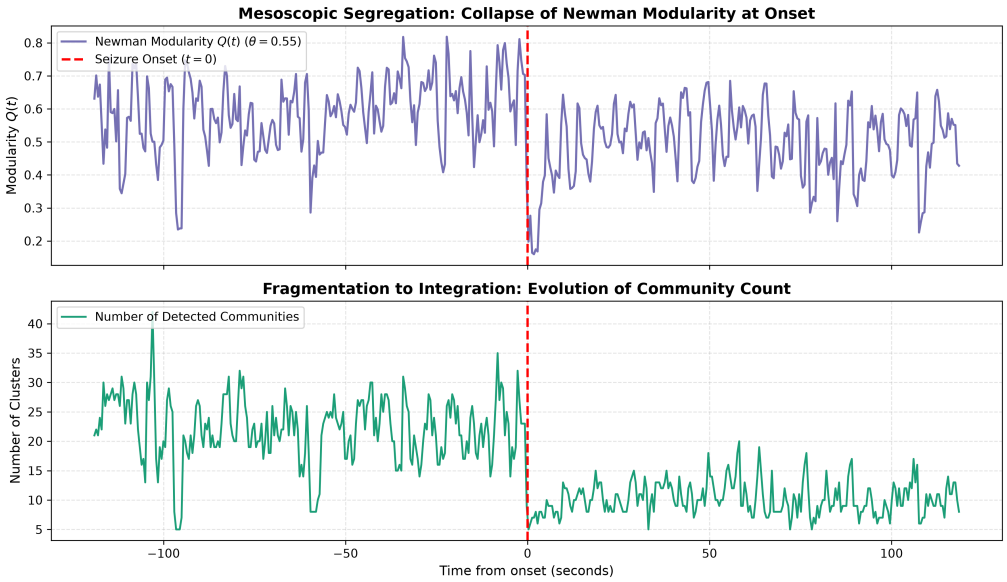


Figure 3.3: Mesoscopic segregation dynamics across seizure onset ($\theta = 0.55$). Top: Evolution of Newman's weighted Modularity $Q(\tau)$, exhibiting a sharp collapse at $t = 0$ s. Bottom: Total number of detected Louvain communities $N_c(\tau)$, illustrating the abrupt transition from pre-ictal fragmentation to ictal integration.

The mesoscopic time series in Figure 3.3 uncover a striking dynamical symmetry breaking that was partially hidden by ceiling effects in the macroscopic order parameter $S(\tau)$:

- **Pre-Ictal Functional Segregation ($t < 0$ s):** Throughout the inter-ictal baseline, the cortical network maintains a highly segregated modular organization, with $Q(\tau)$

fluctuating around a high mean value of $Q \approx 0.60$–0.75 and the graph partitioned into $N_c(\tau) \approx 20$–32 distinct functional clusters. Notably, the two transient inter-ictal bursts previously observed in $\chi(\tau)$ at $\tau \approx -95$ s and $\tau \approx -59$ s appear here as brief, reversible dips where modularity drops ($Q \to 0.24$ and 0.29) and communities momentarily coalesce ($N_c \to 5$ and 8) before immediately rebounding to the segregated baseline.

- **Critical Modularity Collapse at Onset ($t = 0$ s):** Right across the premonitory window ($\tau \in [-15, 0]$ s), modularity reaches elevated values ($Q \approx 0.75$–0.82), reflecting strong local isolation prior to the transition. Exactly at $t = 0$ s, functional segregation collapses catastrophically: $Q(\tau)$ plummets within 2 seconds from ~ 0.80 down to its global minimum of $Q \approx 0.16$, while the number of communities drops precipitously from $N_c \approx 25$–35 down to just $N_c \approx 4$–5 clusters.

- **Post-Onset Ictal Reorganization ($t > +10$ s):** Following the acute onset collapse, the network does not return to its pre-ictal state. Instead, it settles into a new, persistently integrated ictal regime characterized by a permanently reduced community count ($N_c(\tau) \approx 8$–15, roughly half the pre-ictal baseline) and lower average modularity ($Q(\tau) \approx 0.45$–0.55).

To visualize this topological reorganization directly in graph space without relying on anatomical coordinates, Figure 3.4 renders the network at three representative time snapshots using a force-directed Kamada-Kawai spring layout ($\theta = 0.55$), where nodes are colored by their Louvain community assignment.

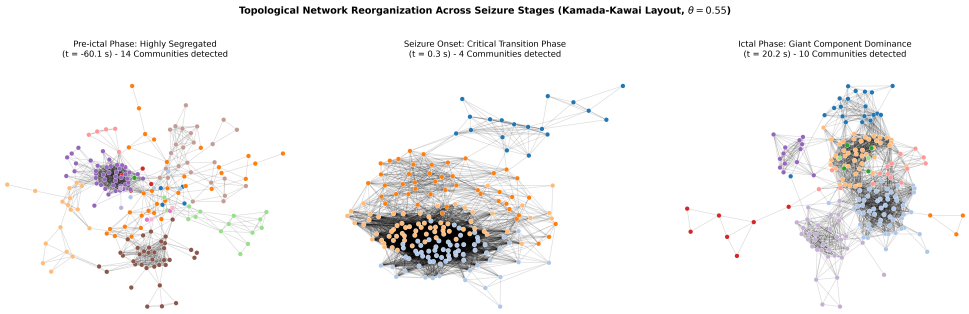


Figure 3.4: Topological network reorganization across seizure stages rendered via force-directed Kamada-Kawai layout ($\theta = 0.55$, isolated singletons omitted for visual clarity). Left: Pre-ictal segregated state ($t = -60.1$ s, 14 active multi-node communities). Center: Seizure onset critical transition ($t = +0.3$ s), where modules collapse into 4 communities dominated by a dense hyper-synchronized core. Right: Established ictal phase ($t = +20.2$ s, 10 communities) showing core-periphery reorganization.

In the pre-ictal snapshot ($t = -60.1$ s, left panel), the network exhibits a distributed web of 14 loosely coupled communities (purple, brown, orange, and green clusters) with sparse inter-modular bridges. At seizure onset ($t = +0.3$ s, center panel), this modular hierarchy

implodes: virtually all nodes are pulled into a massive, tightly bound central syncytium (light-blue and orange nodes) with only 4 communities remaining. By the established ictal phase ($t = +20.2$ s, right panel), the giant component partially differentiates into two main coupled macro-lobes (10 communities total), matching the secondary rebound in $Q(\tau)$ observed in Figure 3.3.

3.2.4 | Critical Assessment and Limitations of the Mesoscopic View

While Newman's Modularity $Q(\tau)$ successfully resolves the internal breakdown of functional segregation at $t = 0$ s, mesoscopic community detection exhibits three methodological limitations that prevent it from answering our clinical question on its own:

1. **Resolution Limit and Degeneracy of Louvain:** Modularity optimization suffers from a well-known resolution limit [3]: when total edge weight $2m(\tau)$ changes drastically across the transition, Louvain may merge small physiological modules or split large ictal clusters unpredictably, and its stochastic node-sweeping order produces slight frame-to-frame jitter in $Q(\tau)$.

2. **Exclusion of Negative Anticorrelations:** Standard Newman modularity (Equation 1.2.6) is formulated strictly for non-negative edge weights $W_{ij}^{+} \geq 0$. Consequently, the mesoscopic view is completely blind to the competitive counter-phase decoupling ($W_{ij} < 0$) that precedes seizure onset.

3. **Nodal Anonymity:** Knowing that $Q(\tau)$ collapses from 0.80 to 0.16 tells us *that* the brain's modular barriers have broken down, but it does not identify *which specific electrode contacts* initiated the decoupling prior to $t = 0$ s or which contacts act as the primary hub drivers of the onset syncytium.

To break nodal anonymity and incorporate the negative correlation spectrum, we must descend to the microscopic scale of individual channels.

3.3 | Microscopic Scale: Premonitory Decoupling and the 2D Functional Role Map

3.3.1 | Physical and Clinical Motivation

Acts I and II demonstrated that seizure onset manifests globally as a critical percolation transition ($S(\tau) \to 1$, $\chi(\tau)$ peak) and mesoscopically as a sudden collapse of modular segregation ($Q(\tau) \to 0.16$). However, from both a dynamical systems and an epileptological perspective, the decisive question lies at the microscopic scale of individual nodes: *which specific sEEG contacts drive this phase transition?*

In particular, guided by the clinical hypothesis of pre-ictal functional detachment [7, 8], we seek to verify whether contacts belonging to the putative epileptogenic network exhibit a premonitory decoupling phase—isolating themselves via negative anticorrelations ($W_{ij} < 0$) in the seconds immediately preceding $t = 0$ s, before recruiting the cortical network into the Giant Connected Component. Furthermore, by pairing pre-ictal decoupling with onset eigenvector centrality across calibrated statistical epochs, we aim to construct an objective two-dimensional classification of nodal roles.

3.3.2 | Numerical Implementation

For each time window τ, we characterize every node $i \in \{1, \dots, N\}$ along two complementary dimensions:

1. **Signed Neighbourhood Correlation and Negative Decoupling:** From the unthresholded signed matrix $\mathbf{W}(\tau)$, we compute each node's mean signed correlation $\bar{W}_i(\tau) = \frac{1}{N-1} \sum_{j \neq i} W_{ij}(\tau)$ to track its net phase alignment, alongside its strictly negative decoupling score $d_i^{(-)}(\tau) = \frac{1}{N-1} \sum_{j \neq i} |\min(0, W_{ij}(\tau))|$ (Equation 1.2.9).

2. **Leading Eigenvector Centrality:** From the cooperative binary adjacency matrix $\mathbf{A}(\tau)$ (thresholded at the calibrated critical value $\theta = 0.55$), we extract the principal eigenvector $\mathbf{v}_1(\tau)$ associated with the largest eigenvalue $\lambda_1(\tau)$ via power iteration, normalized in the L_1-norm ($\sum_i v_{1,i}(\tau) = 1$) so that $v_{1,i}(\tau)$ represents node i's share of total network centrality at frame τ.

To suppress transient inter-ictal noise, nodal roles are evaluated by averaging $d_i^{(-)}(\tau)$ over the Premonitory Window $\mathcal{E}_{\text{prem}} = [-15.0, 0.0]$ s and computing the net centrality surge $\Delta\bar{v}_{1,i} = \langle v_{1,i} \rangle_{\text{onset}} - \langle v_{1,i} \rangle_{\text{base}}$ between the Onset Epoch $\mathcal{E}_{\text{onset}} = [0.0, +15.0]$ s and the calibrated Quiet Baseline Epoch $\mathcal{E}_{\text{base}} = [-84.0, -34.0]$ s:

```
def compute_microscopic_node_roles(W, times, pos_thresh=0.55, epochs=
    None):
    n_wins, n_nodes, _ = W.shape
```

```
    eig_centrality = np.zeros((n_wins, n_nodes), dtype=np.float64)
    neg_decoupling = np.zeros((n_wins, n_nodes), dtype=np.float64)
    mean_signed_corr = np.zeros((n_wins, n_nodes), dtype=np.float64)

    for t in range(n_wins):
        W_t = W[t].copy()
        np.fill_diagonal(W_t, 0.0)

        # Signed mean correlation and negative decoupling strength per node
        mean_signed_corr[t] = np.sum(W_t, axis=1) / (n_nodes - 1)
        neg_only = np.where(W_t < 0.0, -W_t, 0.0)
        neg_decoupling[t] = np.sum(neg_only, axis=1) / (n_nodes - 1)

        # Leading eigenvector centrality v_1(t) on cooperative binary graph A(t)
        adj = (W_t >= pos_thresh).astype(np.float64)
        if np.sum(adj) > 0:
            eigvals, eigvecs = np.linalg.eigh(adj)
            v1 = np.abs(eigvecs[:, -1])  # Principal eigenvector (largest lambda)
            v1_sum = np.sum(v1)
            if v1_sum > 0:
                eig_centrality[t] = v1 / v1_sum  # L1-norm normalization

    # Statistical epoch expectations
    mask_base = (times >= epochs["baseline"][0]) & (times <= epochs["baseline"][1])
    mask_prem = (times >= epochs["premonitory"][0]) & (times <= epochs["premonitory"][1])
    mask_onset = (times >= epochs["onset"][0]) & (times <= epochs["onset"][1])

    premonitory_score = np.mean(neg_decoupling[mask_prem], axis=0)
    delta_v1 = np.mean(eig_centrality[mask_onset], axis=0) - np.mean(eig_centrality[mask_base], axis=0)

    return eig_centrality, mean_signed_corr, premonitory_score, delta_v1
```

Listing 3.3: Microscopic node metrics: L_1-normalized leading eigenvector centrality $\mathbf{v}_1(\tau)$, negative decoupling score $d_i^{(-)}(\tau)$, and statistical epoch aggregation.

3.3.3 | Empirical Results: Premonitory Isolation and the 2D Role Map

Figure 3.5 presents the three-panel microscopic trajectory analysis for recording `DB470450` ($\theta = 0.55$).

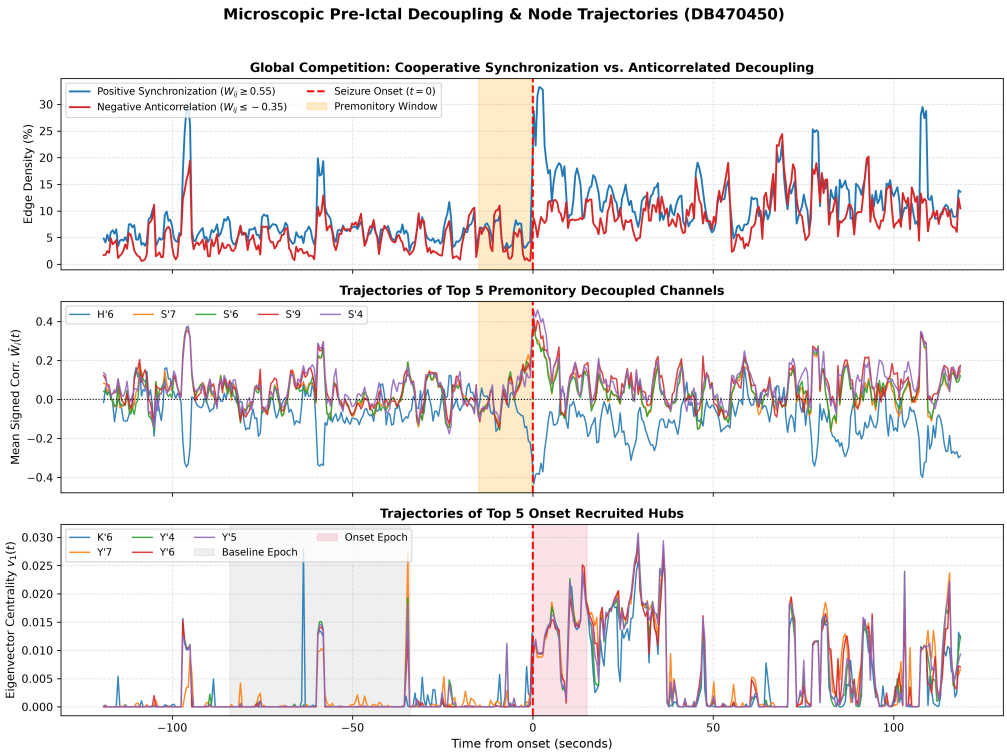


Figure 3.5: Microscopic pre-ictal decoupling and nodal centrality trajectories in recording `DB470450` ($\theta = 0.55$). Top: Global edge density competition between cooperative synchronization ($W_{ij} \geq 0.55$, blue) and negative anticorrelation ($W_{ij} \leq -0.35$, red), highlighting the Premonitory Window ($[-15, 0]$ s, orange shading). Middle: Mean signed correlation trajectories $\bar{W}_i(\tau)$ of the top 5 premonitory decoupled channels (`H'6`, `S'7`, `S'6`, `S'9`, `S'4`). Bottom: Leading eigenvector centrality trajectories $v_{1,i}(\tau)$ of the top 5 onset-recruited hubs (`K'6`, `Y'7`, `Y'4`, `Y'6`, `Y'5`), contrasting the quiet Baseline Epoch ($[-84, -34]$ s, grey shading) against the Onset Epoch ($[0, +15]$ s, pink shading).

Inspecting the middle and bottom panels of Figure 3.5 reveals a remarkable dynamical dichotomy across the critical transition:

- **Premonitory Decoupling and Bifurcation at $t = 0$ s (Middle Panel):** The five channels exhibiting the strongest pre-ictal detachment belong exclusively to two anatomical electrode shafts: contact `H'6` and a contiguous cluster on shaft `S'` (`S'7`, `S'6`, `S'9`, `S'4`). Inside the Premonitory Window ($\tau \in [-15, 0]$ s, orange shading), all five contacts dip into negative mean correlation ($\bar{W}_i(\tau) \approx -0.15$ to -0.20), confirming active functional isolation prior to seizure onset. Crucially, as the system crosses $t = 0$ s, their trajectories undergo a striking bifurcation: the `S'` contacts (`S'4`, `S'6`, `S'7`, `S'9`) rapidly invert their phase and shoot upward to $\bar{W}_i \approx +0.45$, joining the cooperative ictal syncytium; conversely, contact `H'6` (blue trace) plunges sharply downward to $\bar{W}_{\text{H'6}} \approx -0.43$ and remains locked in persistent counter-phase throughout the entire seizure.
- **Onset Centrality Surge and Baseline Robustness (Bottom Panel):** The bottom panel tracks $v_{1,i}(\tau)$ for the five strongest cooperative hubs (`K'6`, `Y'7`, `Y'4`, `Y'6`, `Y'5`). Notice how our automated baseline calibration placed the reference window ($\mathcal{E}_{\text{base}} = [-84, -34]$ s, grey shading) in a predominantly quiet region, avoiding the large inter-ictal burst at $\tau \approx -95$ s and minimizing the impact of the brief spike at $\tau \approx -59$ s. Whereas $v_{1,i}(\tau)$ remains pinned near zero across the baseline, it jumps immediately at $t = 0$ s (pink Onset Epoch) to a sustained high-centrality plateau ($v_{1,i} \in [0.010, 0.030]$ lasting until $\tau \approx +40$ s), identifying shafts `K'` and `Y'` as the core structural backbone of the ictal Giant Connected Component.

To synthesize these two orthogonal dynamical behaviors across the entire array of N electrodes, Figure 3.6 pairs the ranked bar chart of top onset hubs (left panel) with the two-dimensional **Microscopic Node Role Map** (right panel).

The 2D Functional Role Map in Figure 3.6 (right panel) spontaneously partitions the intracranial electrode contacts into four physically and clinically interpretable dynamical regimes:

1. **Premonitory Seizure Initiators (Upper-Right Quadrant, high $\langle d_i^{(-)} \rangle_{\textbf{prem}} > 0.096$, $\Delta\bar{v}_1 > 0$):** Populated by the contiguous shaft `S'` cluster (`S'9`, `S'4`, `S'7`, `S'6`, with decoupling scores reaching 0.115–0.125 and positive centrality gain $\Delta\bar{v}_1 \in [+0.001, +0.007]$). These nodes directly validate the clinical pre-ictal decoupling hypothesis: they actively detach from the physiological network in the 15 seconds prior to onset and subsequently transition into cooperative hubs that help drive the ictal cascade.
2. **Primary Ictal Amplifiers / Core Hubs (Upper-Left Quadrant, low $\langle d_i^{(-)} \rangle_{\textbf{prem}} \approx 0.035$–$0.050$, maximum $\Delta\bar{v}_1 \approx +0.0120$–$0.0124$):** Dominated by a tightly clustered group of contacts along shafts `K'` and `Y'`—led by `K'6` ($\Delta\bar{v}_1 = 0.0124$), followed closely

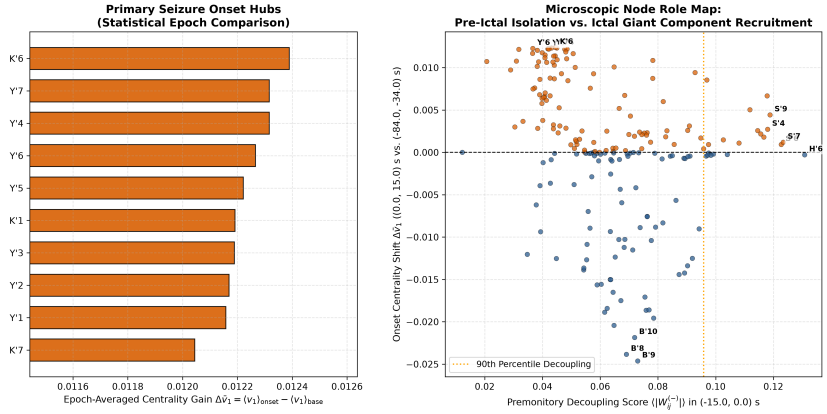


Figure 3.6: Epoch-averaged hub ranking and 2D Microscopic Node Role Map for recording `DB470450`. Left: Top 10 primary seizure onset hubs ranked by epoch-averaged eigenvector centrality gain $\Delta\bar{v}_{1,i} = \langle v_{1,i} \rangle_{\text{onset}} - \langle v_{1,i} \rangle_{\text{base}}$ (horizontal axis rescaled to $[0.0115, 0.0126]$ to resolve fine nodal hierarchy). Right: Joint scatter distribution of all intracranial channels plotting Premonitory Decoupling Score $\langle d_i^{(-)} \rangle_{\text{prem}}$ in $[-15, 0]$ s (x-axis) against Onset Centrality Shift $\Delta\bar{v}_{1,i}$ (y-axis). The vertical orange dotted line marks the 90th percentile of premonitory decoupling; the horizontal black dashed line separates net onset hub recruitment ($\Delta\bar{v}_1 > 0$, orange circles) from centrality suppression ($\Delta\bar{v}_1 < 0$, blue circles).

by `Y'7`, `Y'4`, `Y'6`, `Y'5`, `K'1`, `Y'3`, `Y'2`, `Y'1`, and `K'7` (resolved in the left bar chart). These channels do not exhibit premonitory counter-phase isolation prior to $t = 0$ s; rather, they act as massive cortical resonators that are explosively recruited at $t = 0$ s to form the dominant synchronous core of the Giant Connected Component.

3. **Persistent Counter-Phase Pole (Far-Right Boundary along $\Delta\bar{v}_1 \approx 0$):** Uniquely occupied by contact `H'6`, which achieves the highest premonitory decoupling score in the entire brain ($\langle d_{\text{H'6}}^{(-)} \rangle_{\text{prem}} \approx 0.133$) while maintaining $\Delta\bar{v}_1 \approx 0.000$. Because $\mathbf{v}_1(\tau)$ is computed strictly on cooperative positive links ($W_{ij} \geq 0.55$), `H'6` gains zero positive centrality at onset precisely because it remains locked in strong anti-phase oscillation ($\bar{W}_{\text{H'6}} < 0$) against the Giant Component throughout the seizure.

4. **Demoted Background Nodes (Lower Half-Plane, $\Delta\bar{v}_1 < 0$, blue circles):** Contacts such as `B'9`, `B'8`, and `B'10` ($\Delta\bar{v}_1 \approx -0.021$ to -0.024), which participated in resting-state baseline connectivity during $\mathcal{E}_{\text{base}}$ but lose their relative share of network centrality at $t = 0$ s as the L_1-normalized eigenvector weight concentrates onto the ictal hubs `K'` and `Y'`.

3.3.4 | Critical Assessment and Methodological Caveats of Role Mapping

Although the 2D Functional Role Map provides a clean, data-driven separation between premonitory decoupling channels (`S'`, `H'6`) and onset amplification hubs (`K'`, `Y'`), caution is warranted when interpreting these topological roles in a clinical or predictive context:

- **Retrospective vs. Prospective Prediction:** The Premonitory Window $\mathcal{E}_{\text{prem}} = [-15, 0]$ s is defined retrospectively relative to a known onset $t = 0$ s. Moreover, as visible in the middle panel of Figure 3.5, brief inter-ictal bursts (e.g., at $\tau \approx -95$ s and -59 s) also trigger transient counter-phase deflections in `H'6` and positive spikes in `S'` without evolving into a seizure. Thus, pre-ictal decoupling alone is a necessary dynamical precursor of the transition, but cannot be claimed as a standalone, zero-false-positive prospective seizure predictor without testing over multi-hour inter-ictal recordings.
- **Zero-Lag Correlation Ambiguity:** Because Pearson correlation $W_{ij}(\tau)$ is a symmetric, zero-lag measure ($W_{ij} = W_{ji}$), observing $\bar{W}_{\text{H'6}} < 0$ cannot distinguish whether contact `H'6` is actively generating inhibitory counter-phase output or merely recording the inverted dipole pole of a nearby cortical generator.

3.4 | Cross-Patient Cohort Robustness and Parameter Calibration

A fundamental requirement for any physical model of biological dynamics is generalizability across subjects. While Sections 3.1–3.3 detailed the multiscale cascade on our primary reference recording (`DB470450`), extending the pipeline across the full clinical dataset revealed that raw, uncalibrated topological metrics are highly sensitive to inter-patient experimental variability. Specifically, evaluating the cohort highlighted three sources of heterogeneity that must be controlled to uncover the universal signature of the phase transition:

1. **Discrepancies Between Clinical and Electrophysiological Onset:** In our initial uncalibrated runs, while the `DB` recordings displayed sharp transitions near the nominal metadata timestamp, the remaining recordings exhibited seemingly noisy, uninterpretable pre-ictal profiles. In one recording, the macroscopic surge in synchronization occurred 30–40 s after the annotated marker; in another (e.g., `DQ0040WG`), strong synchronization bursts ($D^+(\tau) > 65\%$) invaded the nominal premonitory window $[-15, 0]$ s well before the annotated $t = 0$ s. Applying the automated step-gradient detector introduced in Equation (2.3.1) resolved this anomaly: clinical metadata often log the delayed behavioral manifestation of the seizure (*clinical onset*) rather than the deep intracranial bifurcation (*electrophysiological onset*). Re-centering $t = 0$ s on the objective energy-synchronization jump restored the sharp step transition across all recordings.

2. **Implantation Density and Sub-Critical Threshold Scaling:** Because each patient receives a personalized stereotactic implantation tailored to their suspected clinical semiology, the spatial density of electrode contacts varies across recordings. In densely sampled implants (such as `DQ0040WG`), applying the default threshold $\theta = 0.50$ resulted in an overly dense baseline graph ($D^+_{\text{base}} \approx 15$–$25\%$), causing frequent spurious spikes in leading eigenvector centrality $v_{1,i}(\tau)$ prior to onset. Scaling the positive threshold $\theta \in [0.50, 0.65]$ via our data-driven criterion—enforcing a sub-critical baseline link density of $D^+_{\text{base}}(\theta) \approx 7\%$—successfully suppressed pre-ictal percolation noise and flattened the baseline centrality traces across the cohort.

3. **Contamination from Extracranial Polygraphic Channels and Pre-Ictal Bursts:** Cross-patient inspection also exposed the critical impact of auxiliary channels and inter-ictal spikes. Prior to implementing the automated label filter (Section 2.1), polygraphic reference channels (such as `POL-7` in recording `DQ0040WG`) consistently ranked among the top 5 premonitory decoupled nodes, driven by subtle pre-ictal muscle or autonomic artifacts that correlated negatively with cortical potentials. Removing `POL`/`ECG` channels and automatically shifting the 50 s Baseline Epoch $\mathcal{E}_{\text{base}}$ to avoid isolated inter-ictal bursts ensured that both the bar rankings and the 2D Role

Maps strictly reflected genuine intracranial neural populations (such as hippocampal `H` and cortical `G`/`D`/`E` shafts).

Once these three objective calibration steps are applied, the qualitative phenomenology of the transition proves remarkably invariant across recordings: every seizure in the cohort is characterized by a sub-critical modular baseline ($Q \approx 0.6$–0.8), a premonitory counter-phase detachment of specific focal contacts in the $[-15, 0]$ s window, and an abrupt percolation collapse at $t = 0$ s where $S(\tau) \to 1$ and $\chi(\tau)$ peaks.

The complete multiscale graphical outputs (Macroscopic Percolation, Mesoscopic Modularity Collapse, and Microscopic Node Role Maps) for the remaining cohort recordings are reported in Appendix B. Across all subjects, the calibrated topological pipeline consistently recovers the transition from pre-ictal modular segregation and focal counter-phase decoupling to global ictal percolation.

4 | Conclusions and Future Perspectives

4.1 | Summary of Findings

In this work, we developed and validated a multiscale Complex Network framework to characterize the pre-ictal to ictal transition in human stereo-electroencephalography (sEEG) recordings. By modeling intracranial recording sites as nodes of a time-varying graph and preserving the algebraic sign of the zero-lag Pearson correlation tensor $\mathbf{W}(\tau) \in [-1, 1]^{N \times N}$, we demonstrated that the onset of a focal epileptic seizure can be rigorously described as a critical phase transition across three hierarchical scales:

- **Macroscopic Scale:** Seizure onset manifests as an abrupt bond percolation transition. When thresholded near the critical boundary ($\theta \in [0.55, 0.65]$), the network transitions discontinuously at $t = 0$ s from a fragmented sub-critical regime into a global Giant Connected Component ($S(\tau) \to 1$). Simultaneously, the spatial variance of the degree distribution—the topological susceptibility $\chi(\tau) = \mathrm{Var}(k(\tau))$—exhibits a sharp, threshold-robust critical peak right at the onset bifurcation.
- **Mesoscopic Scale:** Evaluating Newman's weighted Modularity $Q(\tau)$ via the Louvain algorithm revealed that physiological inter-ictal activity is sustained by a highly segregated architecture ($Q \approx 0.60$–0.80, $N_c \approx 20$–35 communities). At $t = 0$ s, this functional segregation collapses catastrophically ($Q \to 0.16$, $N_c \to 4$), as independent functional modules dissolve into a pathologically integrated syncytium.
- **Microscopic Scale:** By separating cooperative synchronization ($W_{ij} > 0$) from competitive anticorrelation ($W_{ij} < 0$) and averaging nodal observables over calibrated statistical epochs, we confirmed the clinical hypothesis of pre-ictal functional detachment. Mapping each channel's Premonitory Decoupling Score $\langle d_i^{(-)} \rangle_{\mathrm{prem}}$ against its Onset Eigenvector Centrality Gain $\Delta \bar{v}_{1,i}$ yielded a two-dimensional **Functional Role Map** that cleanly separates premonitory seizure initiators (e.g., shaft `S'`), secondary cortical amplifiers (shafts `K'` and `Y'`), and persistent counter-phase poles (`H'6`).

4.2 | Methodological Limitations and Clinical Scope

While the proposed topological descriptors successfully capture the dynamical reorganization of the epileptogenic network, it is essential to frame these results within the proper epistemological boundaries of an exploratory study in Physics of Complex Systems:

1. **Absence of 3D Euclidean Spatial Coordinates:** Throughout this study, the sEEG network was analyzed in a purely functional, topological space without access to the three-dimensional stereotactic coordinates (x_i, y_i, z_i) of the electrode contacts. Consequently, we could not explicitly decouple short-range physiological connectivity driven by spatial proximity and volume conduction from genuine long-range axonal recruitment via an Euclidean distance matrix $D_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$. Nevertheless, the ability of purely topological metrics $(S, \chi, Q, \mathbf{v}_1)$ to isolate contiguous electrode shafts (such as `S'`, `K'`, and `Y'`) demonstrates that intrinsic correlation geometry encodes strong structural information even in the absence of spatial priors.

2. **Retrospective Characterization vs. Prospective Seizure Prediction:** Although our 2D Role Map identifies channels that decouple in the 15 s preceding seizure onset, this analysis is inherently retrospective, as the temporal axis is aligned around a known transition point $t = 0$ s. As observed in our time-series traces, brief inter-ictal epileptiform discharges far from seizure onset (e.g., at $\tau \approx -95$ s and -59 s in `DB470450`) can trigger transient modularity dips, susceptibility spikes, and counter-phase deflections that mimic early onset dynamics before relaxing back to baseline. Claiming these metrics as a standalone, real-time clinical seizure detector or an automated surgical localization tool would require prospective validation over multi-day continuous recordings to quantify false-positive rates and statistical specificity.

3. **Linear Zero-Lag Approximation:** Zero-lag Pearson correlation captures linear, broadband phase locking but is insensitive to nonlinear cross-frequency coupling and directed causal lag. Therefore, distinguishing whether a decoupled node actively inhibits the surrounding network or passively reflects a phase-inverted dipole requires directed spectral estimators.

4.3 | Future Perspectives

The methodological pipeline established in this thesis opens several natural avenues for future investigation at the interface of statistical physics and clinical epileptology:

- **Spatial Embedding and Contact Map Integration:** Incorporating the MNI or native 3D anatomical coordinates of the sEEG contacts will allow the construction of a hybrid spatio-functional tensor. Partitioning functional links by their physical Euclidean length D_{ij} (e.g., short-range intra-shaft links $D_{ij} < 15$ mm vs. long-range inter-areal bridges $D_{ij} \geq 15$ mm) will clarify whether pre-ictal decoupling is a localized micro-circuit phenomenon while ictal percolation is driven by long-range white-matter shortcuts.
- **Frequency-Band Decomposition and Directed Causality:** Filtering the sEEG signals into canonical neurophysiological bands (particularly high-frequency oscillations and fast gamma activity, 80–250 Hz, which are hallmarks of the Seizure Onset Zone) and replacing symmetric Pearson correlation with directed metrics such as Granger Causality or Transfer Entropy could resolve the causal direction of link recruitment in the 2D Role Map.
- **Clinical Validation Against Surgical Outcomes:** Finally, systematically comparing the channels identified in the upper-right (Initiators) and upper-left (Amplifiers) quadrants of our Functional Role Map against clinician-annotated Seizure Onset Zones and post-surgical Engel outcomes across a larger patient cohort will provide a definitive test of the translational value of signed network topology.

Appendices

A | Notation and Conventions

This appendix summarizes the mathematical symbols, signal processing parameters, and graph-theoretical observables adopted throughout the thesis to ensure consistency across the theoretical framework (Chapter 1), the computational implementation (Chapter 2), and the multiscale empirical analysis (Chapter 3).

A.1 | Signal Processing and Temporal Epochs

Continuous stereo-electroencephalography (sEEG) recordings are standardized and segmented around the aligned electrophysiological seizure onset ($t = 0$ s) according to the following conventions:

- $f_s = 1000$ Hz: Sampling frequency of the raw intracranial voltage recordings.
- $x_i(t) \in \mathbb{R}$: Raw electrical potential recorded at intracranial electrode contact i at continuous time t.
- $z_i(t) = (x_i(t) - \mu_i)/\sigma_i$: Channel-wise z-score standardized voltage time series over the extracted 240 s transition chunk ($[-120\text{ s}, +120\text{ s}]$), where μ_i and σ_i denote the temporal mean and standard deviation of channel i.
- $T = 2.0$ s, $\Delta\tau = 0.5$ s: Duration (2000 samples) and step stride (500 samples) of the overlapping sliding windows used for dynamic connectivity estimation, yielding $T_{\text{wins}} = 477$ temporal frames.
- $\tau \in [-119.0\text{ s}, +119.0\text{ s}]$: Discrete time index corresponding to the center of the sliding window $T_\tau = [\tau - T/2, \tau + T/2]$.
- $E_{\text{RMS}}(\tau)$: Root-mean-square signal energy averaged across the top 15% most active channels at window τ, used in conjunction with link density to detect the objective electrophysiological onset τ^*.

- $\mathcal{E}_{\text{base}}, \mathcal{E}_{\text{prem}}, \mathcal{E}_{\text{onset}}, \mathcal{E}_{\text{ictal}}$: Calibrated statistical macro-epochs denoting, respectively, the quiet inter-ictal Baseline Epoch (50 s window prior to -20 s, e.g., $[-84, -34]$ s in `DB470450`), the Premonitory Window ($[-15.0, 0.0]$ s), the Onset Recruitment Epoch ($[0.0, +15.0]$ s), and the Established Ictal Epoch ($[+15.0, +60.0]$ s).

A.2 | Graph Theory and Signed Functional Connectivity

At each temporal frame τ, the cerebral recording array is mapped onto an undirected functional network without spatial coordinates:

- $G(\tau) = (V, E(\tau))$: Time-varying graph defined by the node set $V = \{1, \ldots, N\}$ and edge set $E(\tau)$.
- $N = |V|$: Total number of valid intracranial sEEG recording contacts after filtering out extracranial auxiliary channels (`POL`, `ECG`, `MK`).
- $\mathbf{W}(\tau) \in [-1, 1]^{N \times N}$: Symmetric, zero-diagonal signed functional connectivity matrix at window τ, where $W_{ij}(\tau)$ is the zero-lag sample Pearson correlation coefficient between channels i and j (with $W_{ii}(\tau) = 0$).
- $W_{ij}^{+}(\tau) = \max(0, W_{ij}(\tau))$: Cooperative (in-phase) weighted connectivity projection used for mesoscopic community detection.
- $\theta \in (0, 1)$, $\theta_{\text{neg}} < 0$: Calibrated positive percolation threshold (typically $\theta \in [0.40, 0.65]$, selected to maintain a sub-critical baseline link density of $\sim 7\%$) and negative anticorrelation threshold ($\theta_{\text{neg}} = -0.35$).
- $\mathbf{A}(\tau) \in \{0, 1\}^{N \times N}$: Binary cooperative adjacency matrix obtained via Heaviside thresholding, $A_{ij}(\tau) = \Theta(W_{ij}(\tau) - \theta)$ for $i \neq j$.
- $D^{+}(\tau), D^{-}(\tau) \in [0, 1]$: Global active link densities quantifying the fraction of node pairs satisfying $W_{ij}(\tau) \geq \theta$ (cooperative synchronization) and $W_{ij}(\tau) \leq \theta_{\text{neg}}$ (competitive anticorrelation), respectively.
- $k_i(\tau) = \sum_{j=1}^{N} A_{ij}(\tau)$: Binary degree of node i in the cooperative thresholded graph at time τ.
- $s_i(\tau) = \sum_{j=1}^{N} W_{ij}^{+}(\tau)$: Positive weighted degree (node strength) of contact i, with total network weight $m(\tau) = \frac{1}{2} \sum_{i,j} W_{ij}^{+}(\tau)$.

A.3 | Multiscale Topological Observables

The critical phase transition across $t = 0$ s is tracked across three hierarchical scales using the following order parameters and nodal descriptors:

- $S(\tau) = N_{\mathrm{GCC}}(\tau)/N \in [1/N, 1]$: Macroscopic percolation order parameter, representing the normalized size of the Giant Connected Component (N_{GCC}) of $\mathbf{A}(\tau)$.
- $\chi(\tau) = \mathrm{Var}(k(\tau))$: Topological susceptibility, defined as the spatial variance of the binary node degree distribution $\frac{1}{N}\sum_{i=1}^{N}(k_i(\tau) - \langle k(\tau)\rangle)^2$ across the network at frame τ.
- $Q(\tau) \in [-1/2, 1]$: Newman's weighted modularity, quantifying the degree of mesoscopic functional segregation of partition $\mathcal{C}(\tau)$ relative to a strength-preserving configuration null model.
- $N_c(\tau) = |\mathcal{C}(\tau)|$: Total number of disjoint functional communities detected at window τ via the Louvain heuristic algorithm.
- $\bar{W}_i(\tau) = \frac{1}{N-1}\sum_{j\neq i} W_{ij}(\tau)$: Mean signed correlation of node i with the rest of the cortical array.
- $d_i^{(-)}(\tau) = \frac{1}{N-1}\sum_{j\neq i} |\min(0, W_{ij}(\tau))|$: Instantaneous negative decoupling score of contact i, measuring the average magnitude of its counter-phase anticorrelations, with premonitory expectation $\langle d_i^{(-)}\rangle_{\mathrm{prem}}$ evaluated over $\tau \in [-15, 0]$ s.
- $\mathbf{v}_1(\tau) \in [0, 1]^N$: Leading eigenvector centrality vector of the binary cooperative graph $\mathbf{A}(\tau)$ ($\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$), normalized via the L_1-norm ($\sum_{i=1}^{N} v_{1,i}(\tau) = 1$).
- $\Delta\bar{v}_{1,i} = \langle v_{1,i}\rangle_{\mathrm{onset}} - \langle v_{1,i}\rangle_{\mathrm{base}}$: Net epoch-averaged eigenvector centrality shift between the Onset Epoch and the Baseline Epoch, plotted against $\langle d_i^{(-)}\rangle_{\mathrm{prem}}$ to construct the 2D Microscopic Node Role Map.

B | Supplementary Cohort Analysis

This appendix reports the complete multiscale topological analysis for the additional sEEG recordings in the dataset, obtained after applying extracranial channel filtering (`POL/ECG` removal), electrophysiological onset alignment ($t = 0$ s), and sub-critical baseline threshold calibration.

B.1 | Recording `DB470451`

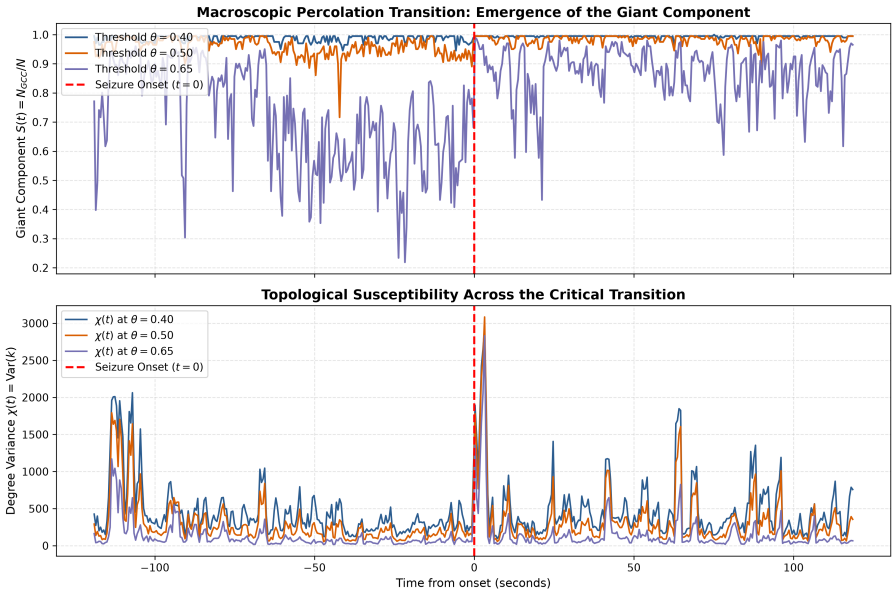


Figure 1: Macroscopic Percolation ($S(\tau)$ and $\chi(\tau)$) for recording `DB470451`.

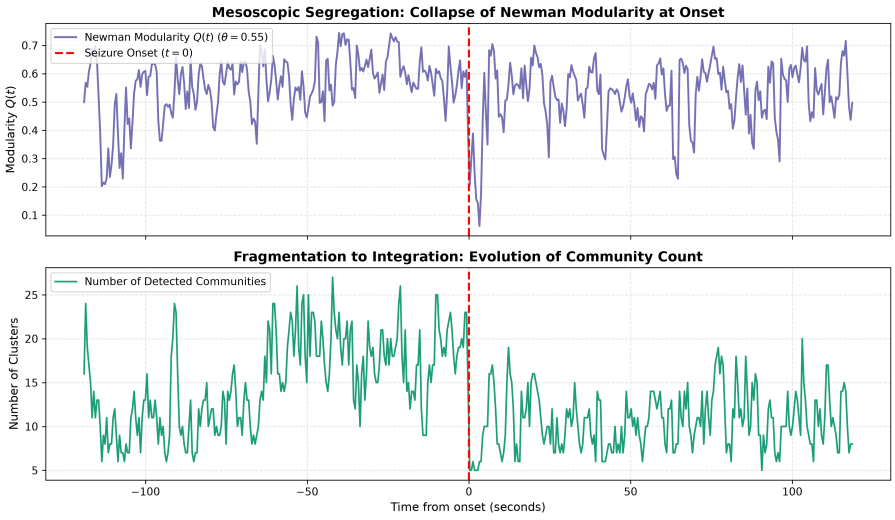


Figure 2: Mesoscopic Modularity Collapse ($Q(\tau)$ and $N_c(\tau)$) for recording `DB470451`.

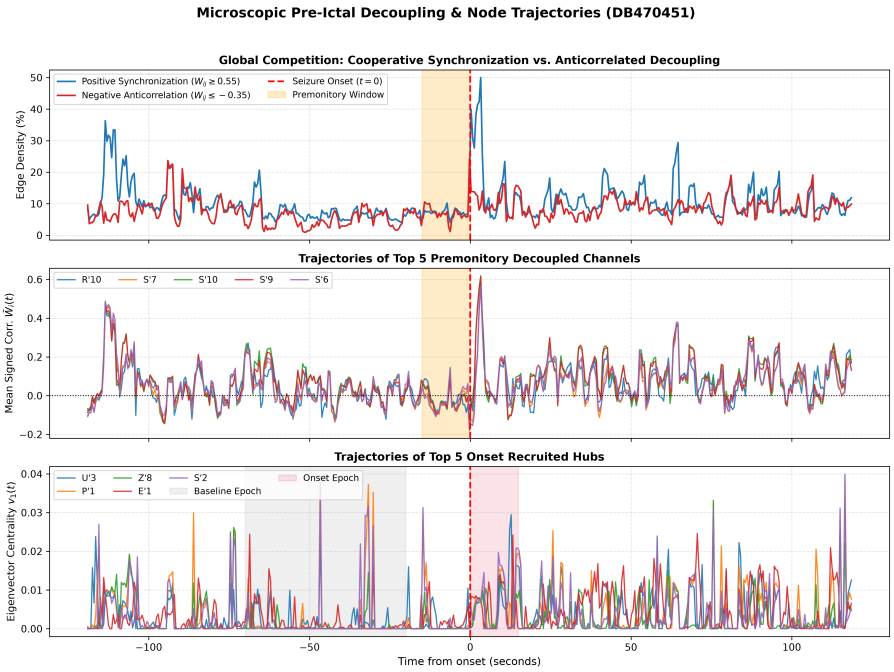


Figure 3: Premonitory Decoupling and Hub Trajectories for recording DB470451.

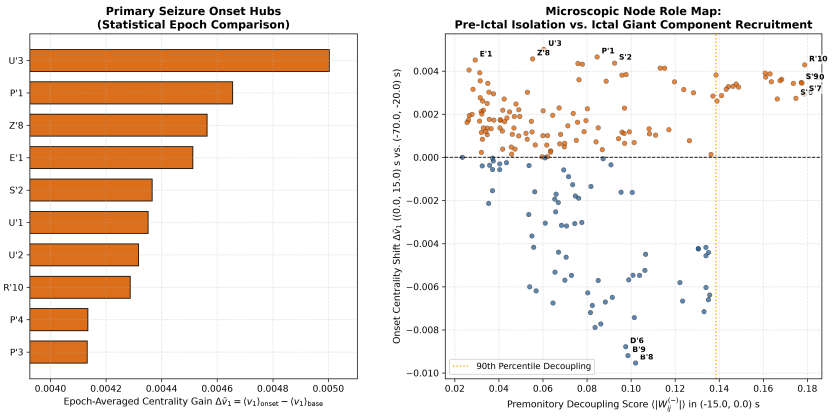


Figure 4: Onset Hub Ranking and 2D Functional Role Map for recording DB470451.

B.2 | Recording DQ0040P6

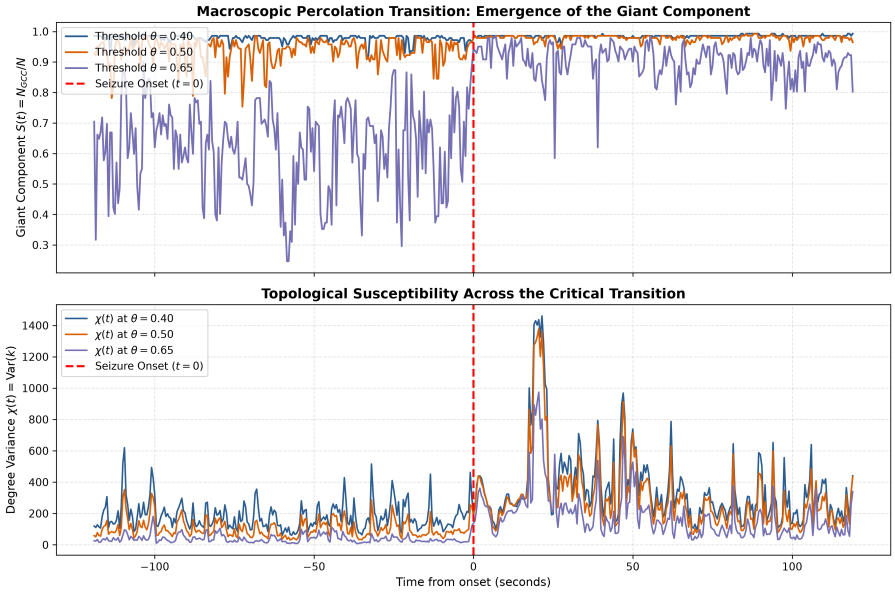


Figure 5: Macroscopic Percolation ($S(\tau)$ and $\chi(\tau)$) for recording DQ0040P6.

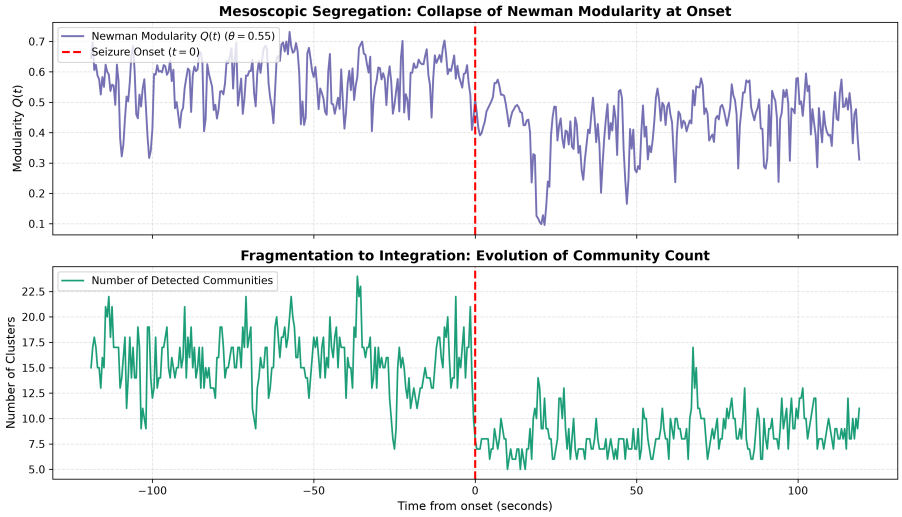


Figure 6: Mesoscopic Modularity Collapse ($Q(\tau)$ and $N_c(\tau)$) for recording DQ0040P6.

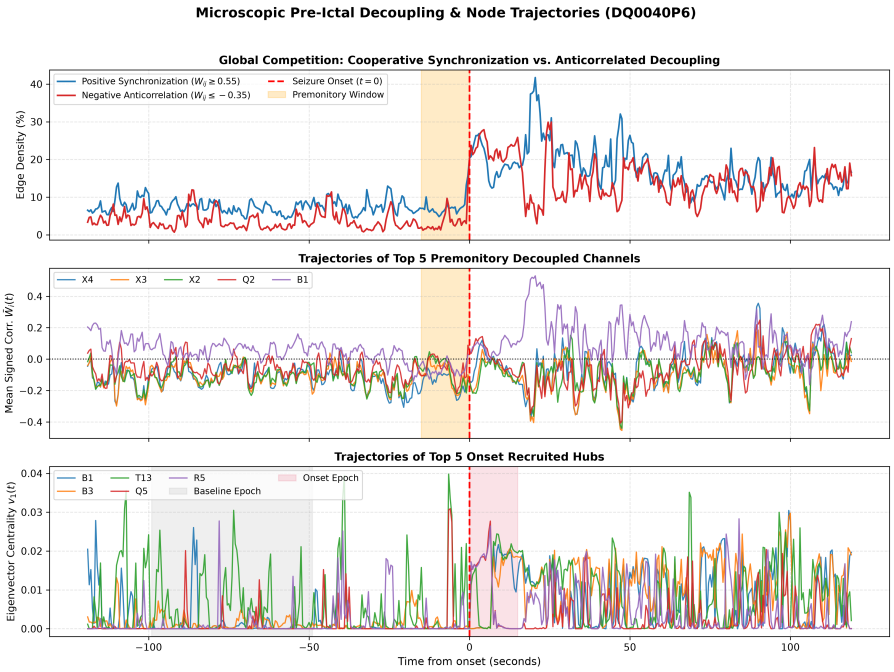


Figure 7: Premonitory Decoupling and Hub Trajectories for recording DQ0040P6.

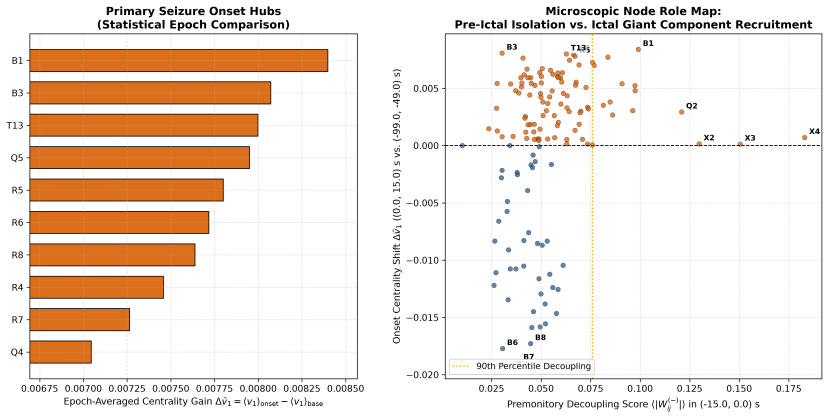


Figure 8: Onset Hub Ranking and 2D Functional Role Map for recording DQ0040P6.

B.3 | Recording DQ0040WG

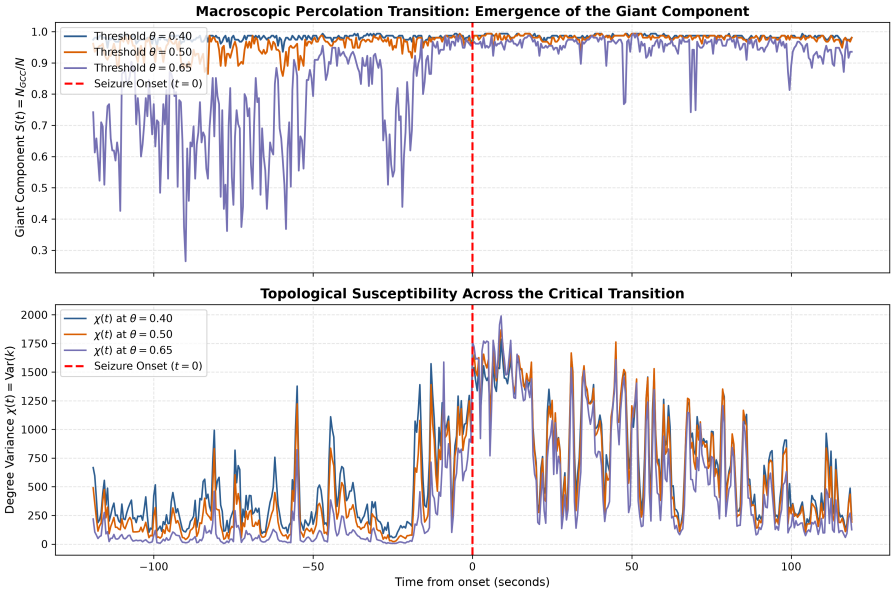


Figure 9: Macroscopic Percolation ($S(\tau)$ and $\chi(\tau)$) for recording DQ0040WG.

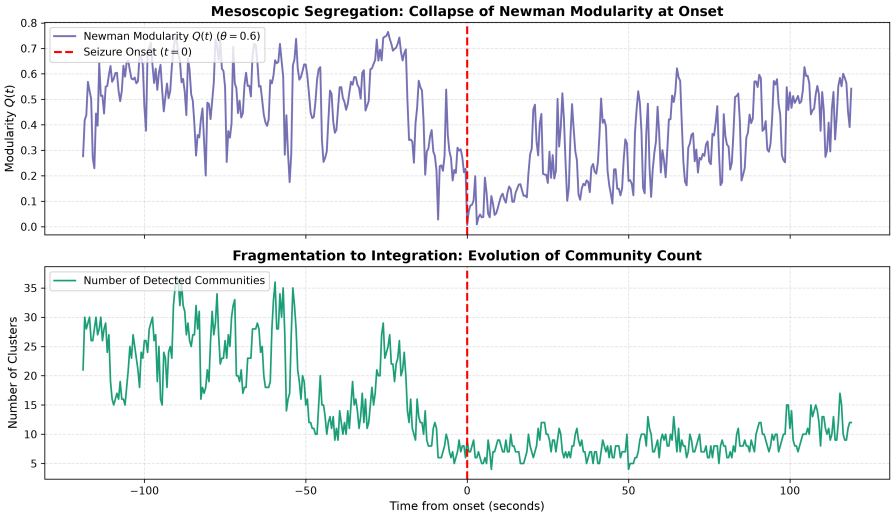


Figure 10: Mesoscopic Modularity Collapse ($Q(\tau)$ and $N_c(\tau)$) for recording DQ0040WG.

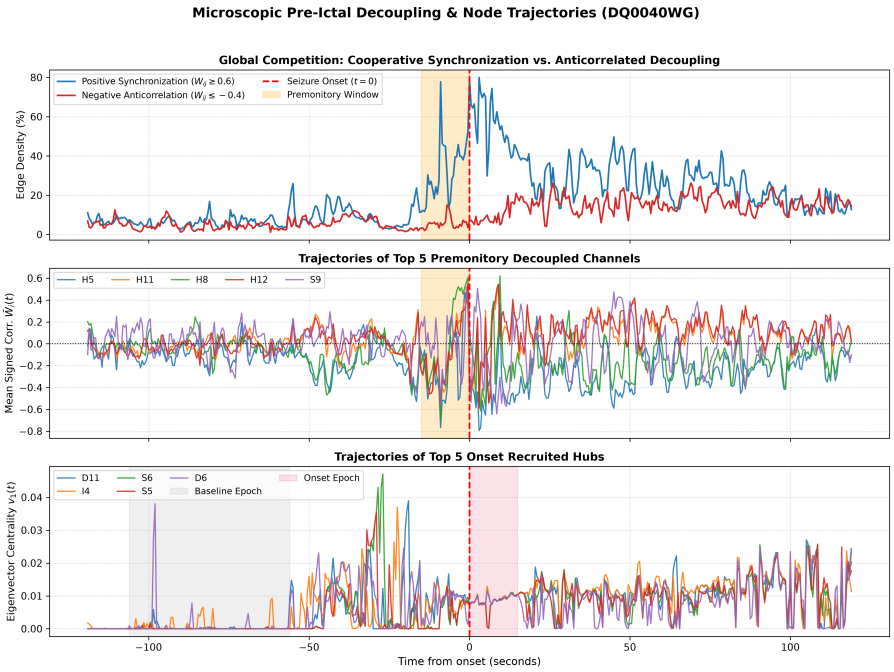


Figure 11: Premonitory Decoupling and Hub Trajectories for recording DQ0040WG.

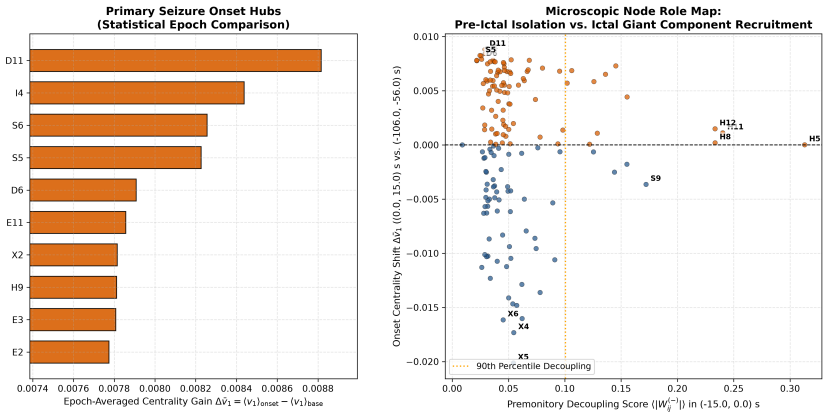


Figure 12: Onset Hub Ranking and 2D Functional Role Map for recording DQ0040WG.

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